QUANTUM SECURITY INTELLIGENCE

Security & Post-Quantum Readiness Audit Report

zebra @ 147a6a8bf7d3 · Zcash Foundation — public repository (passive review)

Organization	QSI
Report date	2026-10-28 13:04 UTC
Report version	1
Audit run	496c9816-0fde-4855-8a56-1a607d666741 · Document audit · 2026-10-28 13:04 UTC
Analyzer engine	qsi-engine 1.2.0
Risk profile	Small technology company
Content SHA-256	`ee7b94a59f12e7f13dc01adc6ec7e927c84e349194b3f02b357fd6092000614f`
Signature key (Ed25519, SHA-256 fingerprint; the signature is the separate file report.sig.json)	`5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`

Contents

1. Executive summary
2. Scope and boundaries
3. Methodology
4. Assets examined and not examined
5. Architecture overview
6. Overall security posture
7. Findings by severity
8. Detailed findings
9. Evidence
10. Post-quantum assessment
11. Limitations and assumptions
12. Remediation plan
13. Priorities for 30, 60 and 90 days
14. Verification after remediation
15. Tools used
16. Sources and normative documents
17. Appendices

This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.

Executive summary

19	0	2	2	9
Findings in total	Critical	High	Medium	Low

Classification	Count
Confirmed technical vulnerability	0
Known vulnerable component	9
Confirmed configuration error	0
Potential cryptographic risk	0
Long-term post-quantum risk	5
Improvement recommendation	4
Insufficient data	1

Top 5

ID	Severity	Detailed findings	Priority score
QSI-2026-05258	High	quinn-proto 0.11.16 is affected by RUSTSEC-2026-0340	38.5
QSI-2026-05259	High	quinn-proto 0.11.16 is affected by RUSTSEC-2026-0341	38.5
QSI-2026-05254	Medium	opentelemetry_sdk 0.31.0 is affected by GHSA-w9wp-h8wv-79jx (CVE-2026-48504)	44.1
QSI-2026-05252	Medium	block-buffer 0.10.4 is affected by GHSA-qwgh-2vcv-g2f7	33.1
QSI-2026-05244	Low	Quantum-vulnerable Ed25519 key (Ed25519) in SECURITY.md	36.2

Mosca's inequality (x + y > z): x = 5, y = 3, z = 12 → time available under this scenario. z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Analyst review: 0 confirmed, 0 not applicable, 1 dismissed as false positives.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography. Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Scope and boundaries

Mode: Document audit

Repository: https://github.com/ZcashFoundation/zebra @ 147a6a8bf7d3b1427a9c0557f284d5bfa7b0e522 — evidence links open the lines at this commit.

Supplied inputs

439 files, 6520513 bytes in total, 0 redaction(s); the per-file SHA-256 list is in appendix C.

Methodology

The audit was performed by the QSI engine 1.2.0. Each analyzer emits facts (rule, parameters, evidence); finding texts are generated from the rule catalog. Methodology sources are listed under "Sources and normative documents".

Prioritization uses an explainable weighted model (severity, exploitability, exposure, confidentiality/integrity/availability impact weighted by the "Small technology company" profile, asset criticality, urgency, ease of remediation) with a confidence multiplier. CVSS v3.1 is applied only to technical vulnerabilities with a defined vector. Long-term quantum risks use Mosca's inequality; their severity is capped at Medium and they are never presented as exploitable vulnerabilities.

Finding classes: confirmed technical vulnerability; known vulnerable component (a dependency version named in an advisory; whether the affected code is reachable is assessed separately, and a component confirmed by review becomes a vulnerability); confirmed configuration error; potential cryptographic risk; long-term post-quantum risk; improvement recommendation; insufficient data. Heuristic findings are marked preliminary; analyst decisions are shown next to the findings.

Coverage of checks

Area	Status	Count	Examples
dependencies	checked	13	Cargo.lock: 709 dependencies parsed (709 pinned); Cargo.toml: 148 dependencies parsed (0 pinned); tower-batch-control/Cargo.toml: 16 dependencies parsed (0 pinned)
keys	Insufficient data	1	SECURITY.md line 70: OpenPGP algorithm ID 18 is not in the algorithm registry
certificates_and_keys	checked	1	SECURITY.md: certificates/keys parsed
source_code	checked	419	book/lens-init.js: 132 lines scanned with 16 rule patterns and 122 registry patterns; book/mermaid-init.js: 1 lines scanned with 16 rule patterns and 122 registry patterns; book/mermaid.min.js: 4 lines scanned with 16 rule patterns and 122 registry patterns
tls_configuration	checked	1	book/src/user/lightwalletd.md: nginx TLS directives parsed
osv_advisories	checked	1	709 pinned packages queried at OSV.dev; 10 affected (2 of them only in development or build dependencies)

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

Assets examined and not examined

Supplied inputs: 435 (input files: C)

Not checked

- keys: SECURITY.md line 70: OpenPGP algorithm ID 18 is not in the algorithm registry

Architecture overview

Static review of the source tree zebra at 147a6a8bf7d3b1427a9c0557f284d5bfa7b0e522. 439 files analysed; of 1600 files, 836 are test, fuzzing, benchmark or example code and were left out, 325 are not read by any analyzer, 0 exceed the per-file limit and 0 did not fit the total limit. No requests were sent to the owner's systems.

Overall security posture

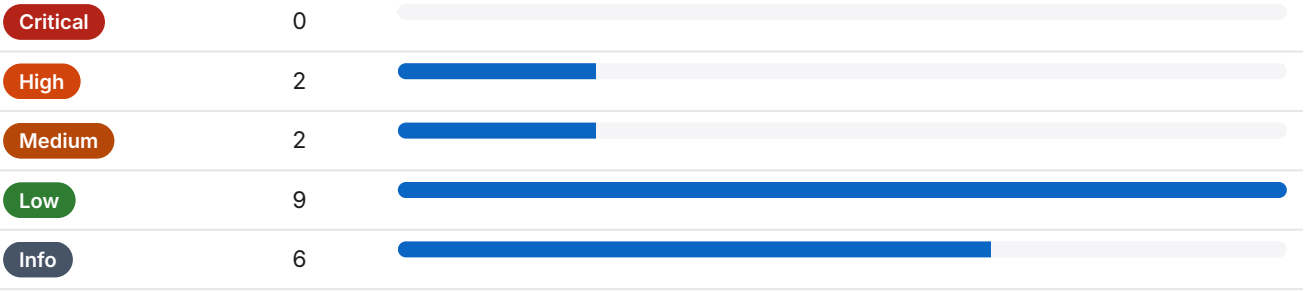
62
Indicative posture index (0–100)

51.9%
Quantum-vulnerable share of inventory

27
Cryptographic inventory

Derived from findings in scope (confirmed and potential issues weighted by severity). Indicative only; not comparable across differently scoped audits.

Findings by severity



ID	Detailed findings	Classification	Severity	Confidence
QSI-2026-05258	quinn-proto 0.11.16 is affected by RUSTSEC-2026-0340	Known vulnerable component	High	Low
QSI-2026-05259	quinn-proto 0.11.16 is affected by RUSTSEC-2026-0341	Known vulnerable component	High	Low
QSI-2026-05254	opentelemetry_sdk 0.31.0 is affected by GHSA-w9wp-h8wv-79jx (CVE-2026-48504)	Known vulnerable component	Medium	Medium
QSI-2026-05252	block-buffer 0.10.4 is affected by GHSA-qwgh-2vcv-g2f7	Known vulnerable component	Medium	Low
QSI-2026-05244	Quantum-vulnerable Ed25519 key (Ed25519) in SECURITY.md	Long-term post-quantum risk	Low	High
QSI-2026-05247	Quantum-vulnerable Ed25519: transaction and message signing (3 file(s))	Long-term post-quantum risk	Low	Medium, preliminary
QSI-2026-05246	Quantum-vulnerable X25519: role not determined (3 file(s))	Long-term post-quantum risk	Low	Medium, preliminary
QSI-2026-05248	Quantum-vulnerable ECDSA: role not determined (1 file(s))	Long-term post-quantum risk	Low	Medium, preliminary
QSI-2026-05249	Quantum-vulnerable BLS12-381: role not determined (2 file(s))	Long-term post-quantum risk	Low	Medium, preliminary
QSI-2026-05256	quinn-proto 0.11.16 is affected by RUSTSEC-2026-0338 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-05257	quinn-proto 0.11.16 is affected by RUSTSEC-2026-0339 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-05260	quinn-proto 0.11.16 is affected by RUSTSEC-2026-0342 (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-05262	webpki-root-certs 1.0.7 is affected by RUSTSEC-2026-0383 (CVE-2026-111943) (no severity rating)	Known vulnerable component	Low	Low, preliminary
QSI-2026-05251	bincode 1.3.3: informational advisory RUSTSEC-2025-0141 (unmaintained)	Improvement recommendation	Info	High
QSI-2026-05261	tokio 1.52.1 is affected by RUSTSEC-2026-0386 (no severity rating)	Known vulnerable component	Info	Low, preliminary

QSI-2026-05263	3 advisories in 2 development or build-only crates.io package(s)	Improvement recommendation	Info	Medium, preliminary
QSI-2026-05250	No post-quantum capable cryptographic library detected	Insufficient data	Info	Medium, preliminary
QSI-2026-05253	number_prefix 0.4.0: informational advisory RUSTSEC-2025-0119 (unmaintained)	Improvement recommendation	Info	Low
QSI-2026-05255	proc-macro-error2 2.0.1: informational advisory RUSTSEC-2026-0173 (unmaintained)	Improvement recommendation	Info	Low

Detailed findings

QSI-2026-05258 · DEP-KNOWN-ADVISORY

High

quinn-proto 0.11.16 is affected by RUSTSEC-2026-0340

Known vulnerable component | Confidence: Low

OSV.dev reports RUSTSEC-2026-0340 for crates.io package quinn-proto at version 0.11.16: Peer minimum ACK delay can panic quinn-proto when ACK_FREQUENCY transmission is enabled. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/quinn-proto@0.11.16
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 13:04 UTC
Priority score	38.5 / 100 (Small technology company)
CVSS v3.1	7.5 — CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0340
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0340 · manifest · SHA-256 20321fd13a708b93...


```
{"id": "RUSTSEC-2026-0340", "aliases": ["GHSA-3g6h-r5qx-f6hq"], "modified":
"2026-10-12T18:15:01.975478256Z", "summary": "Peer minimum ACK delay can panic quinn-proto when
ACK_FREQUENCY transmission is enabled", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/
AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.11.18).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0340

QSI-2026-05259 · DEP-KNOWN-ADVISORY

High

quinn-proto 0.11.16 is affected by RUSTSEC-2026-0341

Known vulnerable component | Confidence: Low

OSV.dev reports RUSTSEC-2026-0341 for crates.io package quinn-proto at version 0.11.16: Stream reassembly memory-exhaustion guard in quinn-proto can be bypassed. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/quinn-proto@0.11.16
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 13:04 UTC
Priority score	38.5 / 100 (Small technology company)
CVSS v3.1	7.5 — CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0341
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0341 · manifest · SHA-256 843224ae2e1a0c89...


```
{"id": "RUSTSEC-2026-0341", "aliases": ["GHSA-qfwj-vfxf-92j2"], "modified":
"2026-10-12T18:15:01.969505433Z", "summary": "Stream reassembly memory-exhaustion guard in quinn-
proto can be bypassed", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/
C:N/I:N/A:H"}], "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.11.17).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0341

QSI-2026-05254 · DEP-KNOWN-ADVISORY

Medium

opentelemetry_sdk 0.31.0 is affected by GHSA-w9wp-h8wv-79jx (CVE-2026-48504)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-w9wp-h8wv-79jx (CVE-2026-48504) for crates.io package opentelemetry_sdk at version 0.31.0: opentelemetry_sdk has unbounded memory allocation in W3C Baggage propagation. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/opentelemetry_sdk@0.31.0
Data source	OSV.dev

Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 13:04 UTC
Priority score	44.1 / 100 (Small technology company)
CVSS v3.1	5.3 — CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:L

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-w9wp-h8wv-79jx
```

EVIDENCE

tool_output · OSV.dev GHSA-w9wp-h8wv-79jx · manifest · SHA-256 2fdd505d93a06e37…


```
{"id": "GHSA-w9wp-h8wv-79jx", "aliases": ["CVE-2026-48504"], "modified":
"2026-09-10T03:50:50.807599062Z", "summary": "opentelemetry_sdk has unbounded memory allocation in
W3C Baggage propagation", "severity": [{"type": "CVSS_V3", "score": "CVSS:3.1/AV:N/AC:L/PR:N/UI:N/
S:U/C:N/I:N/A:L"}], "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (0.32.1).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-w9wp-h8wv-79jx

QSI-2026-05252 · DEP-KNOWN-ADVISORY

Medium

block-buffer 0.10.4 is affected by GHSA-qwgh-2vcv-g2f7

Known vulnerable component | Confidence: Low

OSV.dev reports GHSA-qwgh-2vcv-g2f7 for crates.io package block-buffer at version 0.10.4: block_buffer: panic corrupts inline buffer position. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/block-buffer@0.10.4
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 13:04 UTC
Priority score	33.1 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-qwgh-2vcv-g2f7
```

EVIDENCE

tool_output · OSV.dev GHSA-qwgh-2vcv-g2f7 · manifest · SHA-256 ac22c9ee6301d8c1...


```
{"id": "GHSA-qwgh-2vcv-g2f7", "aliases": [], "modified": "2026-08-19T19:30:06.686866977Z",
"summary": "block_buffer: panic corrupts inline buffer position", "severity": [{"type": "CVSS_V4",
"score": "CVSS:4.0/AV:N/AC:L/AT:P/PR:N/UI:N/VC:L/VI:L/VA:N/SC:N/SI:N/SA:N"}],
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Upgrade to a fixed version (none published yet; see the advisory for mitigations).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-qwgh-2vcv-g2f7

QSI-2026-05244 · KEY-QV-PUBLIC

Low

Quantum-vulnerable Ed25519 key (Ed25519) in SECURITY.md

Long-term post-quantum risk | Confidence: High

A Ed25519 Ed25519 public key was found (line 70).

Affected asset	file:SECURITY.md#line 70
Data source	SECURITY.md
Detection method	Public key parsing.
Analyzer	certificates 1.2.0 · 2026-10-28 13:04 UTC
Priority score	36.2 / 100 (Small technology company)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=5.0, y=3.0, z=12.0 → time available under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 70 in the supplied file SECURITY.md and inspect the quoted lines (see evidence).
```

EVIDENCE

key · SECURITY.md · line 70 · SHA-256 6dbf836a10cb5714... · File fdf8e8c7b66b1c64...

```
OpenPGP primary key: algorithm ID 22 (Ed25519) Ed25519
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-05247 · CODE-QV-ASYMMETRIC

Low

Quantum-vulnerable Ed25519: transaction and message signing (3 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

3 reference(s) to Ed25519 across 3 file(s), e.g. zebra-chain/src/transaction/joinsplit.rs:49. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:Ed25519:transaction_signing
Data source	zebra-chain/src/transaction/joinsplit.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-28 13:04 UTC
Priority score	25.0 / 100 (Small technology company)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=5.0, y=3.0, z=12.0 → time available under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open zebra-chain/src/transaction/joinsplit.rs:49 and inspect the quoted lines; the evidence lists
further files.
```

EVIDENCE

file_excerpt · zebra-chain/src/transaction/joinsplit.rs · line 49 · SHA-256 7b5f18c28d4b4cf9… · File 155b6fd236825920…

```
48 |     /// The JoinSplit signature, denoted as `joinSplitSig` in the spec.
49 |     pub sig: ed25519::Signature,
50 | }
```

file_excerpt · zebra-consensus/src/transaction.rs · line 1340 · SHA-256 c5cd881930df105b… · File cb84a61366b275f7…

```
 1339 |         );
 1340 |         let sig =
zebra_chain::primitives::ed25519::Signature::from(sprout_bundle.joinsplit_sig);
 1341 |         let ed25519_item = (pub_key, sig, shielded_sighash).into();
```

file_excerpt · zebra-rpc/src/methods/types/transaction.rs · line 380 · SHA-256 ad4a7b87509a64c5... · File b0d6338831d0dbd9...

```
  379 |     #[getter(copy)]
  380 |     pub(crate) joinsplit_sig: Option<[u8; ed25519::Signature::BYTE_SIZE]>,
  381 |
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-05246 · CODE-QV-ASYMMETRIC

Low

Quantum-vulnerable X25519: role not determined (3 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

6 reference(s) to X25519 across 3 file(s), e.g. zebra-chain/src/primitives.rs:18. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:X25519:general
Data source	zebra-chain/src/primitives.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-28 13:04 UTC
Priority score	23.6 / 100 (Small technology company)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=5.0, y=3.0, z=12.0 → time available under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open zebra-chain/src/primitives.rs:18 and inspect the quoted lines; the evidence lists further
files.
```

EVIDENCE

file_excerpt · zebra-chain/src/primitives.rs · line 18 · SHA-256 891112a1f260bc48… · File ac08248210a1b071…

```
17 | pub use redjubjub;
18 | pub use x25519_dalek as x25519;
19 |
```

file_excerpt · zebra-chain/src/sprout/arbitrary.rs · line 45 · SHA-256 6cca04a7b460f9f8… · File f60e27780ae48066…

```
44 |                         commitments,
45 |                         ephemeral_key: x25519_dalek::PublicKey::from(ephemeral_key_bytes),
46 |                         random_seed,
```

file_excerpt · zebra-chain/src/sprout/joinsplit.rs · line 11 · SHA-256 04fa2797580674ce… · File 225314a0c4d61733…

```
10 |     fmt::HexDebug,
11 |     primitives::{x25519, Bctv14Proof, Groth16Proof, ZkSnarkProof},
12 |     serialization::{
```

file_excerpt · zebra-chain/src/sprout/joinsplit.rs · line 79 · SHA-256 f6084a25657a411e… · File 225314a0c4d61733…

```
78 |     /// An X25519 public key.
79 |     pub ephemeral_key: x25519::PublicKey,
80 |     /// A 256-bit seed that must be chosen independently at random for each
```

file_excerpt · zebra-chain/src/sprout/joinsplit.rs · line 112 · SHA-256 ed85f11f64c582dc… · File 225314a0c4d61733…

```
111 |     /// An X25519 public key.
112 |     pub ephemeral_key: x25519::PublicKey,
113 |     /// A 256-bit seed that must be chosen independently at random for each
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-05248 · CODE-QV-ASYMMETRIC

Low

Quantum-vulnerable ECDSA: role not determined (1 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

2 reference(s) to ECDSA across 1 file(s), e.g. zebra-chain/src/transparent/keys.rs:8. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:ECDSA:general
Data source	zebra-chain/src/transparent/keys.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-28 13:04 UTC
Priority score	23.6 / 100 (Small technology company)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=5.0, y=3.0, z=12.0 → time available under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open zebra-chain/src/transparent/keys.rs:8 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · zebra-chain/src/transparent/keys.rs · line 8 · SHA-256 e6db1fc498541110… · File 88ba431df453bf5f…

```
   7 |
   8 | use secp256k1::PublicKey;
   9 |
```

file_excerpt · zebra-chain/src/transparent/keys.rs · line 24 · SHA-256 d97661ad177e37ab… · File 88ba431df453bf5f…

```
  23 |         Self::from_slice(&bytes[..])
  24 |             .map_err(|_| SerializationError::Parse("invalid secp256k1 compressed public key"))
  25 |     }
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-05249 · CODE-QV-ASYMMETRIC

Low

Quantum-vulnerable BLS12-381: role not determined (2 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

2 reference(s) to BLS12-381 across 2 file(s), e.g. zebra-consensus/src/primitives/groth16/params.rs:3. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:BLS12-381:general
Data source	zebra-consensus/src/primitives/groth16/params.rs
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-28 13:04 UTC
Priority score	23.6 / 100 (Small technology company)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=5.0, y=3.0, z=12.0 → time available under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open zebra-consensus/src/primitives/groth16/params.rs:3 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · zebra-consensus/src/primitives/groth16/params.rs · line 3 · SHA-256 5e29d8128b44e2bb… · File f0746e2513991205…

```
2 |
3 | use bls12_381::Bls12;
4 | use derive_getters::Getters;
```

file_excerpt · zebra-consensus/src/primitives/groth16.rs · line 6 · SHA-256 9c7451fbee4e424c… · File 278bda853282f1e9…

```
5 | use bellman::{gadgets::multipack, VerificationError};
6 | use bls12_381::Bls12;
7 | use futures::{future::BoxFuture, FutureExt};
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-05256 · DEP-ADVISORY-UNRATED

Low

quinn-proto 0.11.16 is affected by RUSTSEC-2026-0338 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0338 for crates.io package quinn-proto at version 0.11.16: Remote peer can panic quinn-proto with zero-length STREAM frames during unordered reads. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/quinn-proto@0.11.16
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 13:04 UTC
Priority score	20.2 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0338
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0338 · manifest · SHA-256 2d8a96c1ccb5782d...


```
{"id": "RUSTSEC-2026-0338", "aliases": ["GHSA-465w-v9q3-7j98"], "modified":
"2026-10-12T18:15:01.969514645Z", "summary": "Remote peer can panic quinn-proto with zero-length
STREAM frames during unordered reads", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.11.18) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0338

QSI-2026-05257 · DEP-ADVISORY-UNRATED

Low

quinn-proto 0.11.16 is affected by RUSTSEC-2026-0339 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0339 for crates.io package quinn-proto at version 0.11.16: Unbounded growth of pending connection ID retirements in quinn-proto. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/quinn-proto@0.11.16
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 13:04 UTC
Priority score	20.2 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0339
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0339 · manifest · SHA-256 b5c8da254e305100...


```
{"id": "RUSTSEC-2026-0339", "aliases": ["GHSA-hmxj-32vh-65vr"], "modified":
"2026-10-12T18:15:01.956822776Z", "summary": "Unbounded growth of pending connection ID retirements
in quinn-proto", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.11.17) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0339

QSI-2026-05260 · DEP-ADVISORY-UNRATED

Low

quinn-proto 0.11.16 is affected by RUSTSEC-2026-0342 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0342 for crates.io package quinn-proto at version 0.11.16: Zero-length DATAGRAM frames bypass the receive buffer limit in quinn-proto. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/quinn-proto@0.11.16
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 13:04 UTC
Priority score	20.2 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0342
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0342 · manifest · SHA-256 41dfd63166e25757…


```
{"id": "RUSTSEC-2026-0342", "aliases": ["GHSA-2hv7-gw8g-gpq5"], "modified":
"2026-10-12T18:15:01.956681478Z", "summary": "Zero-length DATAGRAM frames bypass the receive buffer
limit in quinn-proto", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.11.17) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0342

QSI-2026-05262 · DEP-ADVISORY-UNRATED

Low

webpki-root-certs 1.0.7 is affected by RUSTSEC-2026-0383 (CVE-2026-111943) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0383 (CVE-2026-111943) for crates.io package webpki-root-certs at version 1.0.7: Bundled root certificates allow trust in distrusted emSign intermediates. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. The advisory names no affected functions, so reachability was not determined.

Affected asset	dependency:crates.io/webpki-root-certs@1.0.7

Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 13:04 UTC
Priority score	20.2 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0383
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0383 · manifest · SHA-256 5020317ed72c7e0c...


```
{"id": "RUSTSEC-2026-0383", "aliases": ["CVE-2026-111943", "GO-2026-6809", "RUSTSEC-2026-0381",
"RUSTSEC-2026-0382"], "modified": "2026-10-21T06:30:02.691648061Z", "summary": "Bundled root
certificates allow trust in distrusted emSign intermediates", "severity": null,
"declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.26.11, 1.0.10) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0383

QSI-2026-05251 · DEP-ADVISORY-INFORMATIONAL

Info

bincode 1.3.3: informational advisory RUSTSEC-2025-0141 (unmaintained)

Improvement recommendation | Confidence: High

OSV.dev lists RUSTSEC-2025-0141 for crates.io package bincode at version 1.3.3 as informational (unmaintained): Bincode is unmaintained. It reports no vulnerability; an unmaintained package will not receive fixes for future ones.

Affected asset	dependency:crates.io/bincode@1.3.3
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 13:04 UTC
Priority score	20.4 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

No known vulnerability. Future security fixes may not be released.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2025-0141
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2025-0141 · manifest · SHA-256 2d4ec91d374dc1d1...


```
{"id": "RUSTSEC-2025-0141", "aliases": [], "modified": "2026-01-16T11:07:28.867553Z", "summary":
"Bincode is unmaintained", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time.

RECOMMENDATION

Plan a move to a maintained alternative, or record the decision to keep it.

RETEST PROCEDURE

Re-run the dependency scan after the change.

OSV advisory RUSTSEC-2025-0141

QSI-2026-05261 · DEP-ADVISORY-UNRATED

Info

tokio 1.52.1 is affected by RUSTSEC-2026-0386 (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports RUSTSEC-2026-0386 for crates.io package tokio at version 1.52.1: `AsyncFd::new` and related constructors violate I/O safety. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. None of the affected functions (tokio::io::unix::AsyncFd::new, tokio::io::unix::AsyncFd::try_new, tokio::io::unix::AsyncFd::try_with_interest, tokio::io::unix::AsyncFd::with_interest) is named in the audited code: the advisory does not apply unless they are reached indirectly (traits, macros, other dependencies).

Affected asset	dependency:crates.io/tokio@1.52.1
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 13:04 UTC
Priority score	17.1 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0386
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0386 · manifest · SHA-256 1ebb594fd9923dc0...


```
{"id": "RUSTSEC-2026-0386", "aliases": [], "modified": "2026-10-23T14:45:03.847864944Z", "summary":
"`AsyncFd::new` and related constructors violate I/O safety", "severity": null,
"declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (1.51.6, 1.53.3) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory RUSTSEC-2026-0386

QSI-2026-05263 · DEP-ADVISORY-DEV

Info

3 advisories in 2 development or build-only crates.io package(s)

Improvement recommendation | Confidence: Medium | preliminary — requires further verification

OSV.dev lists 3 advisories (1 rated high or critical) for 2 crates.io package(s) that only development or build tooling pulls in (Cargo.lock): e.g. git2@0.20.4 [build]: RUSTSEC-2026-0183, RUSTSEC-2026-0184; pulldown-cmark@0.13.3 [build]: RUSTSEC-2026-0414. The manifests mark them as such (devDependencies, "dev": true in package-lock.json, pnpm importers sections, [dev-dependencies]/[build-dependencies] reached through Cargo.lock, or — for yarn.lock, which records no scope at all — package.json reached through the lock graph), so they are not part of what ships.

Affected asset	dependencies:crates.io:non-runtime
Data source	OSV.dev
Detection method	OSV.dev querybatch lookup of exact versions; scope from devDependencies, package-lock.json dev flags, pnpm importers sections, the Cargo.lock graph rooted at the Cargo.toml tables, and — for yarn.lock, which records no scope — the lock graph walked from package.json. Where a runtime requirement cannot be resolved in a lock, that lock is left unclassified rather than risking a package being set aside wrongly.
Analyzer	dependencies 1.2.0 · 2026-10-28 13:04 UTC
Priority score	16.0 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

They do not reach users of the product. They can affect developer machines and CI (supply chain): a compromised or vulnerable tool can tamper with builds.

REPRODUCTION STEPS

```
Look the packages listed in the evidence up at https://osv.dev and check their scope in Cargo.lock.
```

EVIDENCE

tool_output · OSV.dev (crates.io) · Cargo.lock · SHA-256 ef4e8ed1d4a904a9...

```
git2@0.20.4 [build]: RUSTSEC-2026-0183, RUSTSEC-2026-0184
pulldown-cmark@0.13.3 [build]: RUSTSEC-2026-0414
```

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Scopes come from the manifests; a registry crate's own build dependencies cannot be told apart in Cargo.lock, so the runtime set may be larger than reality, never smaller.

RECOMMENDATION

Update the tooling during routine maintenance; prioritise packages that run in CI with secrets or that produce release artefacts.

RETEST PROCEDURE

Re-run the dependency scan after updating.

OSV.dev — open-source vulnerability database and API

QSI-2026-05250 · DEP-NO-PQC-LIBRARY

Info

No post-quantum capable cryptographic library detected

Insufficient data | Confidence: Medium | preliminary — requires further verification

Across 13 dependency manifests, QSI found crypto libraries (curve25519-dalek, ring, rustls, secp256k1, x25519-dalek) but none from its PQC-capable list.

Affected asset	dependencies
Data source	dependency manifests
Detection method	Dependency name matching.
Analyzer	dependencies 1.2.0 · 2026-10-28 13:04 UTC
Priority score	13.4 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Readiness indicator only: PQC support may come from the platform (OpenSSL, OS, cloud KMS) rather than application dependencies.

REPRODUCTION STEPS

```
Review the dependency manifests listed in the evidence.
```

EVIDENCE

dependency · Cargo.lock, Cargo.toml, tower-batch-control/Cargo.toml, tower-fallback/Cargo.toml, zebra-chain/Cargo.toml, zebra-consensus/Cargo.toml, zebra-network/Cargo.toml, zebra-node-services/Cargo.toml, zebra-rpc/Cargo.toml, zebra-script/Cargo.toml, zebra-state/Cargo.toml, zebra-utils/Cargo.toml, zebrad/Cargo.toml · SHA-256 b2f21b0b5cbbd4c5...

```
Crypto libraries: curve25519-dalek, ring, rustls, secp256k1, x25519-dalek
```

LIMITATIONS OF THE CHECK

Absence of evidence is not evidence of absence: system libraries and transitive dependencies may not be listed.

RECOMMENDATION

Confirm where PQC support will come from (TLS terminator, KMS/HSM, library upgrade).

RETEST PROCEDURE

Re-scan after upgrades.

FIPS 203 — Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM); FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA)

QSI-2026-05253 · DEP-ADVISORY-INFORMATIONAL

Info

number_prefix 0.4.0: informational advisory RUSTSEC-2025-0119 (unmaintained)

Improvement recommendation | Confidence: Low

OSV.dev lists RUSTSEC-2025-0119 for crates.io package number_prefix at version 0.4.0 as informational (unmaintained): number_prefix crate is unmaintained. It reports no vulnerability; an unmaintained package will not receive fixes for future ones.

Affected asset	dependency:crates.io/number_prefix@0.4.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 13:04 UTC
Priority score	12.3 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

No known vulnerability. Future security fixes may not be released.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2025-0119
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2025-0119 · manifest · SHA-256 fdb2df9598c08223...


```
{"id": "RUSTSEC-2025-0119", "aliases": [], "modified": "2025-11-18T08:13:56Z", "summary":
"number_prefix crate is unmaintained", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time.

RECOMMENDATION

Plan a move to a maintained alternative, or record the decision to keep it.

RETEST PROCEDURE

Re-run the dependency scan after the change.

OSV advisory RUSTSEC-2025-0119

QSI-2026-05255 · DEP-ADVISORY-INFORMATIONAL

Info

proc-macro-error2 2.0.1: informational advisory RUSTSEC-2026-0173 (unmaintained)

Improvement recommendation | Confidence: Low

OSV.dev lists RUSTSEC-2026-0173 for crates.io package proc-macro-error2 at version 2.0.1 as informational (unmaintained): proc-macro-error2 is unmaintained. It reports no vulnerability; an unmaintained package will not receive fixes for future ones.

Affected asset	dependency:crates.io/proc-macro-error2@2.0.1
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.2.0 · 2026-10-28 13:04 UTC
Priority score	12.3 / 100 (Small technology company)
CVSS v3.1	not applicable

POTENTIAL IMPACT

No known vulnerability. Future security fixes may not be released.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/RUSTSEC-2026-0173
```

EVIDENCE

tool_output · OSV.dev RUSTSEC-2026-0173 · manifest · SHA-256 0ecc182141bd9eff...


```
{"id": "RUSTSEC-2026-0173", "aliases": [], "modified": "2026-06-07T21:00:04.318567692Z", "summary":
"proc-macro-error2 is unmaintained", "severity": null, "declared_in_manifest": false}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time.

RECOMMENDATION

Plan a move to a maintained alternative, or record the decision to keep it.

RETEST PROCEDURE

Re-run the dependency scan after the change.

OSV advisory RUSTSEC-2026-0173

Dismissed after review

The analyst dismissed these findings as false positives; they are not counted in this report.

ID	Detailed findings	Analyst review
QSI-2026-05245	Non-cryptographic RNG near security-sensitive identifiers in book/mermaid.min.js	book/mermaid.min.js is the minified Mermaid diagram library bundled into the mdBook documentation output. It is third-party code vendored for rendering docs, never runs in the node, and the Math.random calls are Mermaid's own layout code rather than anything security-sensitive in Zebra.

Evidence

ID	Type	Data source	SHA-256
QSI-2026-05258	tool_output	OSV.dev RUSTSEC-2026-0340 (manifest)	20321fd13a708b936ab6ccc115a6ed97f9b78eb43a4e76d134ec1664a5fc8
QSI-2026-05259	tool_output	OSV.dev RUSTSEC-2026-0341 (manifest)	843224ae2e1a0c898b003d66aa1610a8f10928c662ad0f000421250465a63
QSI-2026-05254	tool_output	OSV.dev GHSA-w9wp-h8wv-79jx (manifest)	2fdd505d93a06e3777486ba943c941dd9227d6544f3653f928247551ce338
QSI-2026-05252	tool_output	OSV.dev GHSA-qwgh-2vcv-g2f7 (manifest)	ac22c9ee6301d8c1e4b09ca6951303d71527ccba2df30c96af92531d8efbb
QSI-2026-05244	key	SECURITY.md (line 70)	6dbf836a10cb57142effcc3f89dd5e74340870bac059dd57e9f528bf344e8
QSI-2026-05247	file_excerpt	zebra-chain/src/transaction/joinsplit.rs (line 49)	7b5f18c28d4b4cf995c8c645d6699ea435fc2e3771326fafe98976648f3dc
QSI-2026-05247	file_excerpt	zebra-consensus/src/transaction.rs (line 1340)	c5cd881930df105bad00840a473fa63cec1e81b86bb4c8096b356109c586e
QSI-2026-05247	file_excerpt	zebra-rpc/src/methods/types/transaction.rs (line 380)	ad4a7b87509a64c51a6849f560fe90747a31fde83bee6d59e22d8cff6b3eb
QSI-2026-05246	file_excerpt	zebra-chain/src/primitives.rs (line 18)	891112a1f260bc480aedb877c789e052eb787dd7a112114ee4a3e84515a36
QSI-2026-05246	file_excerpt	zebra-chain/src/sprout/arbitrary.rs (line 45)	6cca04a7b460f9f82202ad7ad023ebc45568a78ab720f6ad0d67cbbfcdceb
QSI-2026-05246	file_excerpt	zebra-chain/src/sprout/joinsplit.rs (line 11)	04fa2797580674ce63171073ee35b73933dbee77849cf3418329cec5339ca
QSI-2026-05246	file_excerpt	zebra-chain/src/sprout/joinsplit.rs (line 79)	f6084a25657a411e99c200c5a6172e4c4dcc30f2a2bece2efc2ba74e614b0
QSI-2026-05246	file_excerpt	zebra-chain/src/sprout/joinsplit.rs (line 112)	ed85f11f64c582dc33a2adabf9830c2e07e0075ff6d95e607338500956ae1
QSI-2026-05248	file_excerpt	zebra-chain/src/transparent/keys.rs (line 8)	e6db1fc498541110585d827e4bac54be3b4b372fdaeff6e13618a2acdab2c

QSI-2026-05248	file_excerpt	zebra-chain/src/transparent/keys.rs (line 24)	d97661ad177e37ab994807da754d0fe72a88b080e505a55e17bef046f0255
QSI-2026-05249	file_excerpt	zebra-consensus/src/primitives/groth16/params.rs (line 3)	5e29d8128b44e2bb4efd6dc475ca29362aab72f84889d77e9d20bdf3b044b
QSI-2026-05249	file_excerpt	zebra-consensus/src/primitives/groth16.rs (line 6)	9c7451fbee4e424ccf8cc8e4eab9f659bc9f37487c42750508c0b205aff30
QSI-2026-05256	tool_output	OSV.dev RUSTSEC-2026-0338 (manifest)	2d8a96c1ccb5782dfd51ab6c6bfc548092df0a416eb5a7c29ca5b5ffdce60
QSI-2026-05257	tool_output	OSV.dev RUSTSEC-2026-0339 (manifest)	b5c8da254e3051004bc694e865ed6cf27e63c10cddc5641f13f42bd290bdb
QSI-2026-05260	tool_output	OSV.dev RUSTSEC-2026-0342 (manifest)	41dfd63166e25757cdb189c219a98c5e4b8d6b2e389f484cc583556a37807
QSI-2026-05262	tool_output	OSV.dev RUSTSEC-2026-0383 (manifest)	5020317ed72c7e0cbf2d22efcc385ee146b42c336ffbeb3bf453e64cec1e9
QSI-2026-05251	tool_output	OSV.dev RUSTSEC-2025-0141 (manifest)	2d4ec91d374dc1d163952a86f9da46edd4e89ca1a0fd828fd523b2b1dcfe3
QSI-2026-05261	tool_output	OSV.dev RUSTSEC-2026-0386 (manifest)	1ebb594fd9923dc03209b41738d3f9cf8f324941e8844709473c643cd247e
QSI-2026-05263	tool_output	OSV.dev (crates.io) (Cargo.lock)	ef4e8ed1d4a904a99effd30e97246700ef1f5b6684029bd2111f907ae591a
QSI-2026-05250	dependency	Cargo.lock, Cargo.toml, tower-batch-control/Cargo.toml, tower-fallback/Cargo.toml, zebra-chain/Cargo.toml, zebra-consensus/Cargo.toml, zebra-network/Cargo.toml, zebra-node-services/Cargo.toml, zebra-rpc/Cargo.toml, zebra-script/Cargo.toml, zebra-state/Cargo.toml, zebra-utils/Cargo.toml, zebrad/Cargo.toml	b2f21b0b5cbbd4c5768559961a4aea5f6ec3bcb12da3000819a21669637b5

QSI-2026-05253	tool_output	OSV.dev RUSTSEC-2025-0119 (manifest)	`fdb2df9598c08223c197071bbc0a302c79f15f783a40297c48b4513e4162c`
QSI-2026-05255	tool_output	OSV.dev RUSTSEC-2026-0173 (manifest)	`0ecc182141bd9effc6d35bc24dc20c57e6473fb3877ad59a5fbdb1cd55b56`

Post-quantum assessment

	Key establishment (x = how long data must stay confidential; harvest now, decrypt later)	Signatures (x = how long signatures/ keys must stay trusted)
Data/key lifetime x	5	5
Migration time y	3	3
Assumed years to CRQC z	12 (central)	12
x + y − z	-4 → time available under this scenario	-4 → time available under this scenario

No signature lifetime is set for the project; the data lifetime is used as an assumption.

Planning assumptions: x for key establishment, x for signatures, y are the defaults of the risk profile “Small technology company”, not the client's figures; with the client's actual lifetimes the verdict can change.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Levels: 0 none; 1 present, not wired; 2 in use outside the critical paths; 3 hybrid in a critical path; 4 post-quantum by default in the critical paths (root of trust, key wrapping, channel key exchange). Judged from code and manifests.

What the quantum-vulnerable algorithms do

Role	Algorithms (files)	ID	First step
transaction and message signing	Ed25519 (3)	QSI-2026-05247	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them
role not determined	X25519 (3), BLS12-381 (2), ECDSA (1)	QSI-2026-05246, QSI-2026-05248, QSI-2026-05249	Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH; introduce an abstraction layer for crypto-agility

Roles are judged from file paths, function names and the code around each match; review confirms them. Root-of-trust keys and wrapped keys come first: they are hard to replace or can be recorded today.

z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Quantum status	Count
Grover: acceptable to NIST, below ECCG/ANSSI guidance	3
n/a	10
vulnerable (Shor)	14

Detailed findings: QSI-2026-05244, QSI-2026-05247, QSI-2026-05246, QSI-2026-05248, QSI-2026-05249

PQC readiness questionnaire

No readiness questionnaire was completed for this project.

Quantum algorithms relevant to this inventory

Quantum algorithm	Effect	Inventoried algorithms (rows)	What it means for this audit
Shor's algorithm — discrete logarithms in finite fields Q1: quantum computation offline, classical access to the victim	breaks	BLS12-381 (2), Groth16 (2)	Breaks finite-field Diffie–Hellman and DSA at every size. Move key establishment to ML-KEM (hybrid with X25519 or ECDH during the transition) and signatures to ML-DSA or SLH-DSA.
Shor's algorithm — elliptic-curve discrete logarithms Q1: quantum computation offline, classical access to the victim	breaks	BLS12-381 (2), ECDSA (1), Ed25519 (5), Groth16 (2), X25519 (4)	Elliptic-curve keys are expected to fall before RSA keys of comparable classical strength. On blockchains, coins whose public key is already on chain (reused addresses, pay-to-public-key and Taproot outputs) are exposed first; plan the move to post-quantum signatures and avoid key reuse in the meantime.
Grover's algorithm — key search and preimages Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	SHA-2 (3)	Not a break. 256-bit keys (AES-256, ChaCha20) keep a comfortable margin; AES-128 meets NIST category 1, while some authorities ask for 256-bit keys in quantum-resistant contexts (see the AES-128 entry). Prefer 256-bit keys for long-lived data; 256-bit hashes need no change for preimage resistance.
Brassard–Høyer–Tapp — collision search Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	SHA-2 (3)	256-bit hashes (SHA-256, SHA3-256) remain adequate for collision resistance; 160-bit hashes are already weak classically (SHA-1 collisions are practical). Use 384/512-bit hashes where NIST categories 4–5 are required.

Effects: breaks — replace the algorithm; weakens — larger keys or outputs restore the margin; included in standard parameters — no action at standard parameter sets; only with superposition queries — not applicable to ordinary deployments.

Limitations and assumptions

- This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.
- z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.
- keys — Insufficient data: SECURITY.md line 70: OpenPGP algorithm ID 18 is not in the algorithm registry

Remediation plan

QSI-2026-05258 · P2 · 30 days

root-cause fix

Upgrade quinn-proto to a fixed version (0.11.18).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0340.

QSI-2026-05259 · P2 · 30 days

root-cause fix

Upgrade quinn-proto to a fixed version (0.11.17).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0341.

QSI-2026-05254 · P2 · 30 days

root-cause fix

Upgrade opentelemetry_sdk to a fixed version (0.32.1).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-w9wp-h8wv-79jx (CVE-2026-48504).

QSI-2026-05252 · P2 · 30 days

root-cause fix

Upgrade block-buffer to a fixed version (none published yet; see the advisory for mitigations).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.

Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-qwgh-2vcv-g2f7.

QSI-2026-05256 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0338; upgrade quinn-proto to a fixed version (0.11.18) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0338, or the triage is recorded.

QSI-2026-05257 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0339; upgrade quinn-proto to a fixed version (0.11.17) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0339, or the triage is recorded.

QSI-2026-05260 · P3 · 60 days root-cause fix

Triage RUSTSEC-2026-0342; upgrade quinn-proto to a fixed version (0.11.17) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.

Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0342, or the triage is recorded.

QSI-2026-05262 · P3 · 60 days

root-cause fix

Triage RUSTSEC-2026-0383 (CVE-2026-111943); upgrade webpki-root-certs to a fixed version (0.26.11, 1.0.10) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0383 (CVE-2026-111943), or the triage is recorded.

QSI-2026-05261 · P3 · 60 days

root-cause fix

Triage RUSTSEC-2026-0386; upgrade tokio to a fixed version (1.51.6, 1.53.3) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports RUSTSEC-2026-0386, or the triage is recorded.

QSI-2026-05244 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at file:SECURITY.md to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-05247 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-05246 · P3 · 90 days

short-term risk reduction

Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-05248 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-05249 · P3 · 90 days

short-term risk reduction

Add BLS12-381 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-05244 · P4 · 90 days

root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-05247 · P4 · 90 days

root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.

Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-05246 · P4 · 90 days

root-cause fix

Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH or a hybrid scheme in use.

QSI-2026-05248 · P4 · 90 days

root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-05249 · P4 · 90 days

root-cause fix

Plan migration of BLS12-381 to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.

Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows or a hybrid scheme in use.

QSI-2026-05251 · P4 · 90 days

root-cause fix

Replace bincode with a maintained alternative or record why it stays.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	medium
Dependencies	Change window and owner approval.
Side effects	Code changes may be required.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	bincode replaced, or the decision recorded.

QSI-2026-05263 · P4 · 90 days

root-cause fix

Update the development and build dependencies listed in the evidence.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Tooling upgrades may change build output; compare release artefacts.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	The dependency scan reports no advisories for tooling packages, or the remaining ones are recorded.

QSI-2026-05253 · P4 · 90 days

root-cause fix

Replace number_prefix with a maintained alternative or record why it stays.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	medium
Dependencies	Change window and owner approval.
Side effects	Code changes may be required.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	number_prefix replaced, or the decision recorded.

QSI-2026-05255 · P4 · 90 days

root-cause fix

Replace proc-macro-error2 with a maintained alternative or record why it stays.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	medium
Dependencies	Change window and owner approval.
Side effects	Code changes may be required.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	proc-macro-error2 replaced, or the decision recorded.

Priorities for 30, 60 and 90 days

30 days

Priority	ID	Action	Type
P2	QSI-2026-05258	Upgrade quinn-proto to a fixed version (0.11.18).	root-cause fix
P2	QSI-2026-05259	Upgrade quinn-proto to a fixed version (0.11.17).	root-cause fix
P2	QSI-2026-05254	Upgrade opentelemetry_sdk to a fixed version (0.32.1).	root-cause fix
P2	QSI-2026-05252	Upgrade block-buffer to a fixed version (none published yet; see the advisory for mitigations).	root-cause fix

60 days

Priority	ID	Action	Type
P3	QSI-2026-05256	Triage RUSTSEC-2026-0338; upgrade quinn-proto to a fixed version (0.11.18) if it applies.	root-cause fix
P3	QSI-2026-05257	Triage RUSTSEC-2026-0339; upgrade quinn-proto to a fixed version (0.11.17) if it applies.	root-cause fix
P3	QSI-2026-05260	Triage RUSTSEC-2026-0342; upgrade quinn-proto to a fixed version (0.11.17) if it applies.	root-cause fix
P3	QSI-2026-05262	Triage RUSTSEC-2026-0383 (CVE-2026-111943); upgrade webpki-root-certs to a fixed version (0.26.11, 1.0.10) if it applies.	root-cause fix
P3	QSI-2026-05261	Triage RUSTSEC-2026-0386; upgrade tokio to a fixed version (1.51.6, 1.53.3) if it applies.	root-cause fix

90 days

Priority	ID	Action	Type
P3	QSI-2026-05244	Add Ed25519 usages at file:SECURITY.md to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-05247	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-05246	Add X25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-05248	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction

P3	QSI-2026-05249	Add BLS12-381 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P4	QSI-2026-05244	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-05247	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-05246	Plan migration of X25519 to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-05248	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-05249	Plan migration of BLS12-381 to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-05251	Replace bincode with a maintained alternative or record why it stays.	root-cause fix
P4	QSI-2026-05263	Update the development and build dependencies listed in the evidence.	root-cause fix
P4	QSI-2026-05253	Replace number_prefix with a maintained alternative or record why it stays.	root-cause fix
P4	QSI-2026-05255	Replace proc-macro-error2 with a maintained alternative or record why it stays.	root-cause fix

Verification after remediation

ID	Retest procedure
QSI-2026-05258	Re-run the dependency scan after upgrading.
QSI-2026-05259	Re-run the dependency scan after upgrading.
QSI-2026-05254	Re-run the dependency scan after upgrading.
QSI-2026-05252	Re-run the dependency scan after upgrading.
QSI-2026-05244	Re-audit after migration.
QSI-2026-05247	Re-run the inventory after refactoring.
QSI-2026-05246	Re-run the inventory after refactoring.
QSI-2026-05248	Re-run the inventory after refactoring.
QSI-2026-05249	Re-run the inventory after refactoring.
QSI-2026-05256	Re-run the dependency scan after upgrading.
QSI-2026-05257	Re-run the dependency scan after upgrading.
QSI-2026-05260	Re-run the dependency scan after upgrading.
QSI-2026-05262	Re-run the dependency scan after upgrading.
QSI-2026-05251	Re-run the dependency scan after the change.
QSI-2026-05261	Re-run the dependency scan after upgrading.
QSI-2026-05263	Re-run the dependency scan after updating.
QSI-2026-05250	Re-scan after upgrades.
QSI-2026-05253	Re-run the dependency scan after the change.
QSI-2026-05255	Re-run the dependency scan after the change.

Tools used

Tool	Version
blockchain	1.0.0
certificates	1.1.0
chain_map	1.0.0
configs	1.1.0
dependencies	1.2.0
inventory_document	1.1.0
solidity	1.2.0
source_code	1.3.0
tls_code	1.1.0
tls_live	1.0.0
qsi-engine	1.2.0
OSV.dev API	v1

Sources and normative documents

Reviewed: 2026-10-13

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5. Common Vulnerability Scoring System v3.1: Specification Document — FIRST. https://www.first.org/cvss/v3.1/specification-document (final)
6. D. J. Bernstein — Cost analysis of hash collisions: Will quantum computers make SHARCS obsolete? (2009) — cr.yp.to. https://cr.yp.to/hash/collisioncost-20090823.pdf (paper dated 2009-08-23)
7. D. Litinski — How to compute a 256-bit elliptic curve private key with only 50 million Toffoli gates (arXiv:2306.08585) — arXiv. https://arxiv.org/abs/2306.08585 (preprint (2023-06-14): 6000 modules × 1152 physical qubits, one key per 10 minutes under its architecture assumptions)
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9. FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA) — NIST. https://csrc.nist.gov/pubs/fips/204/final (final (August 2024))
10. FIPS 205 — Stateless Hash-Based Digital Signature Standard (SLH-DSA) — NIST. https://csrc.nist.gov/pubs/fips/205/final (final (August 2024))
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12. J. Proos, C. Zalka — Shor's discrete logarithm quantum algorithm for elliptic curves (Quantum Inf. Comput. 3(4), 2003; arXiv:quant-ph/0301141) — Quantum Information & Computation. https://doi.org/10.26421/QIC3.4-3 (peer-reviewed article)
13. L. K. Grover — A fast quantum mechanical algorithm for database search (STOC 1996) — ACM. https://doi.org/10.1145/237814.237866 (peer-reviewed article)
14. M. Amy et al. — Estimating the cost of generic quantum pre-image attacks on SHA-2 and SHA-3 (SAC 2016; arXiv:1603.09383) — Springer / arXiv. https://arxiv.org/abs/1603.09383 (peer-reviewed article: SHA-256 about 2^153.8 surface-code cycles on 2^12.6 logical qubits; SHA3-256 about 2^146.5 cycles on 2^20)
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18. M. Roetteler, M. Naehrig, K. Svore, K. Lauter — Quantum resource estimates for computing elliptic curve discrete logarithms (ASIACRYPT 2017) — IACR / arXiv. https://arxiv.org/abs/1706.06752 (peer-reviewed article: at most 9n + 2⌈log2 n⌉ + 10 qubits and 448 n^3 log2 n + 4090 n^3 Toffoli gates)
19. NIST IR 8547 — Transition to Post-Quantum Cryptography Standards — NIST. https://csrc.nist.gov/pubs/ir/8547/ipd (initial public draft (November 2024); proposes deprecating quantum-vulnerable public-key algorithms after 2030 and disallowing them after 2035 — dates may change in the final version)
20. NIST — Post-Quantum Cryptography Standardization: Evaluation Criteria, Security (Call for Proposals, Section 4.A.5) — NIST. https://csrc.nist.gov/Projects/Post-Quantum-Cryptography/Post-Quantum-Cryptography-Standardization/Evaluation-Criteria/Security-(Evaluation-Criteria) (security categories: AES-128 key search 2^170/MAXDEPTH quantum gates, AES-192 2^233/MAXDEPTH, AES-256 2^298/MAXDEPTH; MAXDEPTH 2^40–2^96)

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36. S. Jaques, M. Naehrig, M. Roetteler, F. Virdia — Implementing Grover oracles for quantum key search on AES and LowMC (EUROCRYPT 2020; arXiv:1910.01700) — Springer / arXiv. https://arxiv.org/abs/1910.01700 (peer-reviewed article: lower cost estimates for the AES key searches that define the NIST categories)

Appendices

A. Cryptographic inventory

Algorithm	Parameter	Location	Quantum status	Classical status
SHA-2	SHA-256	zebra-chain/src/serialization/sha256d.rs:17	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
SHA-2	SHA-256	zebra-chain/src/sprout/commitment.rs:53	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
SHA-2	SHA-256	zebrad/src/components/zcashd_compat/managed.rs:424	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable
Crypto library ring (ring (Rust))	0.17.14	Cargo.lock	n/a	acceptable
Crypto library rustls (rustls)	0.23.46	Cargo.lock	n/a	acceptable
Crypto library secp256k1 (secp256k1 bindings)	0.33.1	Cargo.lock	n/a	acceptable
Crypto library x25519-dalek (ed25519-dalek)	3.0.0	Cargo.lock	n/a	acceptable
Crypto library curve25519-dalek (ed25519-dalek)	5.0.0	Cargo.lock	n/a	acceptable
Crypto library secp256k1 (secp256k1 bindings)	0.33	Cargo.toml	n/a	acceptable
Crypto library x25519-dalek (ed25519-dalek)	3.0.0	Cargo.toml	n/a	acceptable
Crypto library secp256k1 (secp256k1 bindings)	workspace	zebra-chain/Cargo.toml	n/a	acceptable
Crypto library x25519-dalek (ed25519-dalek)	workspace	zebra-chain/Cargo.toml	n/a	acceptable
Crypto library secp256k1 (secp256k1 bindings)	workspace	zebra-script/Cargo.toml	n/a	acceptable
BLS12-381		zebra-consensus/src/primitives/groth16.rs:6	vulnerable (Shor)	acceptable
BLS12-381		zebra-consensus/src/primitives/groth16/params.rs:3	vulnerable (Shor)	acceptable

ECDSA	secp256k1	zebra-chain/src/transparent/keys.rs:8	vulnerable (Shor)	acceptable
Ed25519	Ed25519	SECURITY.md (line 70)	vulnerable (Shor)	acceptable
Ed25519	Ed25519	zebra-chain/src/transaction/joinsplit.rs:49	vulnerable (Shor)	acceptable
Ed25519	Ed25519	zebra-consensus/src/primitives/ed25519.rs	vulnerable (Shor)	acceptable
Ed25519	Ed25519	zebra-consensus/src/transaction.rs:1340	vulnerable (Shor)	acceptable
Ed25519	Ed25519	zebra-rpc/src/methods/types/transaction.rs:380	vulnerable (Shor)	acceptable
Groth16		zebra-chain/src/primitives/proofs/groth16.rs	vulnerable (Shor)	acceptable
Groth16		zebra-consensus/src/primitives/groth16.rs	vulnerable (Shor)	acceptable
X25519	X25519	zebra-chain/src/primitives.rs:18	vulnerable (Shor)	acceptable
X25519	X25519	zebra-chain/src/sprout/arbitrary.rs:45	vulnerable (Shor)	acceptable
X25519	X25519	zebra-chain/src/sprout/joinsplit.rs:11	vulnerable (Shor)	acceptable
X25519	X25519	zebra-chain/src/sprout/joinsplit.rs:220	vulnerable (Shor)	acceptable

B. JSON

qsi.report/1 · SHA-256 ee7b94a59f12e7f13dc01adc6ec7e927c84e349194b3f02b357fd6092000614f

C. Input files

File	Size	SHA-256	Re
.cargo/config.toml	603	`dff3518ee1e2ad6ee6398cfa26a9fd7848684707097382aa5fc6b9f89a16055f`	0
Cargo.lock	187356	`31268fbfed2d1ab05e8e561c75b7ed13f252cc361b8cabb4ddb2d6bc1f828a48`	0
Cargo.toml	9681	`f40661a9b68a8e78079cd541f480fd8607b99585587d198d8a0251e70478a854`	0
SECURITY.md	7370	`fdf8e8c7b66b1c640102ef75de09b95e23577560eec69998771454639f621771`	0
book/lens-init.js	5043	`49002bf2f55877de5dd757cf6edba104327a2872d379a9961e2b8784a1207f79`	0
book/mermaid-init.js	40	`4533fb1122e7a9ba074cd643af9dfcb80ac55d856f404809b64f825a968a69b1`	0
book/mermaid.min.js	1102136	`cc85eceaea4013d34222f27899942954cf0bab6ea5bb905d5e729c863f0d2bab`	0
book/src/user/lightwalletd.md	12256	`ad8a9013e04f614e75d6ea060f2b8a14392de076d6e7cf4f4a600fd9762c6ea2`	0
rust-toolchain.toml	116	`cdfe339851a0498ade3c898923a8b85ef0d0b8e62f89b17acedd84215f631b39`	0
tower-batch-control/Cargo.toml	1803	`628573df8b60fb6427ea3bddb62ecf9c590dd11b9fc83c0aa40dc4d43d053d58`	0
tower-batch-control/src/error.rs	1481	`35901dc0c75beb0414425cde78b7db648f67d2084b0d7abbc3f0641f72a3097b`	0

tower-batch-control/src/future.rs	2114	b2f05b6976b9ef7337cd5f919d945a010d12de0107c6015a20725999155bf4f5	0
tower-batch-control/src/layer.rs	2302	db41eec5e5d5da7bce0a24984dc3e734575c56991b3cf835ba3ef49553e6d028	0
tower-batch-control/src/lib.rs	7087	e78ad738c7495b0fc8ef038be6c6c581e3a8ae1a5ab29fab2c3637fa06e01a7c	0
tower-batch-control/src/message.rs	537	9ecd17807122051b4d73c1cae2ada043d94e2197bf90fbd5519000ceab431daf	0
tower-batch-control/src/service.rs	13398	0f9c9a11cac0c21849ee2a30521da26758aafc2e173d1fda4b44a42ce1fde5a0	0
tower-batch-control/src/worker.rs	15068	d5f5c91bc6ca570dfeeca60931872783b802cccb2cf7bf2f81c63134eab7e77a	0
tower-fallback/Cargo.toml	966	6f6d497e2823096f5c843614281abd4a20b45c814cfef72d5c6bb451ddccfb96	0
tower-fallback/src/future.rs	6118	821fa33ac72bbc373655ac60bec7d1bdb9bb12d7b242c47c16b082e525976e8f	0
tower-fallback/src/lib.rs	894	57c7730d3834b8872b4e0c63997aa705e0b74b346c5a97449342efde5aef3cd4	0
tower-fallback/src/service.rs	3021	43782a22f8f7dd114b81d5cfa25160fb8365d27ab7121dc826e4318bcdb1f2eb	0
zebra-chain/Cargo.toml	4688	44e15e5b237327f489597cda1b77c69c440ac927d8551c96bb9e8fd5ae517cf1	0
zebra-chain/src/amount/arbitrary.rs	389	aa32760cc0157fc35a503adbe2b568cc17bf1eaae0a97723d40147f10748dd9d	0
zebra-chain/src/amount.rs	18625	e15a28102acccd42d64392c3bcceefaf9a8faf5d454f12b12ed68360060378e0	0
zebra-chain/src/block/arbitrary.rs	34669	098a8c1d268edd40a80dc7f36ba78a97fd5e045f6c10a76419ff51da21fad9d3	0
zebra-chain/src/block/commitment.rs	14574	39b46bb7447e89451c4f526ff658eaec8d40bcb52b28dbfa71fb615d0fd1d69c	0
zebra-chain/src/block/error.rs	296	a4b8243c60f155e0034561d8f001f711e2b6199984582b318ea15ce337d384b3	0
zebra-chain/src/block/genesis.rs	632	4b9028a0c6e7d6a1572d9626d80f7ff968ed984ab2e8b6d34108bc3ac41dfa21	0
zebra-chain/src/block/hash.rs	3859	ca00b4348b24744c4601a13d9e6957770718f3420b2fb245877fffff3f27b8a8	0
zebra-chain/src/block/header.rs	7709	f39aaff28500b78710c17b046dba65d696145ab1912d6f1f2aa901048a4abd24	0
zebra-chain/src/block/height/json_conversion.rs	610	1fa73aa793ecf9186042e1c2f8e087f1c22cdb405a5bd6d7a38504fadeb9683e	0
zebra-chain/src/block/height.rs	11056	6c660e1495d45b55f90dcedd12197e98f4e630b2146fffdcaca65a29d8822b5d	0
zebra-chain/src/block/merkle.rs	14822	407c6eac0137b4b323f42dd2bcab9c2718b95f153ad302e298e8cc8c961d929e	0
zebra-chain/src/block/serialize.rs	8112	029e7606cc75087e0a1db4b51c6d0735b4ce31953da2ca14072e6eac016331b6	0
zebra-chain/src/block.rs	17661	9bd39dc802c8dc796514c3a8e56868edd9b4542100ab6c6591abba776a2d2431	0
zebra-chain/src/block_info.rs	818	4beefa42b13e2c438e0428d87809b183dea12853b6962a72f4124931794e8358	0
zebra-chain/src/chain_sync_status/mock.rs	731	36a21322f4b5a99a0093e89ee05dde7b99ddd210a86528b996f18d0c858998fe	0
zebra-chain/src/chain_sync_status.rs	485	b95881d999d704f40e1d96ab5839c2771e222ccbbddd883cc0b90a92cad6f318	0
zebra-chain/src/chain_tip/mock.rs	6939	ee59ec57f1eb77fb0333f0e5a470ea7b1e392c01e5a7ae7cb553bc0c68072c45	0
zebra-chain/src/chain_tip/network_chain_tip_height_estimator.rs	6483	1be19e70741e76dd84e206e4169961ef18571480ecb81ca890da1bc26bff23c9	0
zebra-chain/src/chain_tip.rs	6495	8097d0688ae51b012415e5d78b861d4fdf0a0d95e1f17a4a6654749d574adefb	0
zebra-chain/src/common.rs	2131	dcf90854986f510c261e7814bf9abe77777bb5fd076366311635200bf534e89f	0

zebra-chain/src/diagnostic/task/future.rs	3908	d955b3670e3125abde50c6f96687ec5ab0dc11bf2c88702266c7271d06be45cb	0
zebra-chain/src/diagnostic/task/thread.rs	4841	e247c84fbdf62349aa9d9a3e7c19203bef587fe1d7b2616fb3edb281585791a5	0
zebra-chain/src/diagnostic/task.rs	3711	0265858a459153dcb9620469bd90a74910ffebbf3346aa42338d00206085ac05	0
zebra-chain/src/diagnostic.rs	4205	63bd5393a2157e454c91162a51d0f0d67a6436c2fa6ebd31dfb30f00261cfeb6	0
zebra-chain/src/error.rs	3996	4afde6617d6b3ee1dcb4d884f113c2000ac9e5fb485141df04885cb28eb75956	0
zebra-chain/src/fmt/time.rs	2342	1649f9d937e997da7d23a08efcc5263447c16147812462da5386198971515831	0
zebra-chain/src/fmt.rs	5555	4da4e06d8c9649a4a2f1a4a6878891242641e652563688aebf11c6bfc9a48b1f	0
zebra-chain/src/history_tree.rs	19397	0cb24cc368a7a17db48a674a001cbe091531adc30fc3db321db2a4d736b75bfd	0
zebra-chain/src/ironwood/arbitrary.rs	1233	043fd48b3c50d09bf4763c2d903590f1931d9c2ee4d24e3c5ad49dfdc6131bee	0
zebra-chain/src/ironwood.rs	2953	de87d15629219937b55b53f685b037d1aeb748534c685990a6d56e6d07806ebf	0
zebra-chain/src/lib.rs	1562	e48b0ec19655441fe086c03c49c784dbb43a84163576de607cddbe8c073d7c4e	0
zebra-chain/src/orchard/action.rs	5420	c5068a760712065f44b778c4ccdb4533bb457d78b4b9d6011a68ebf13a1922aa	0
zebra-chain/src/orchard/address.rs	820	257fc309060e5636708c22dc699845f5dcef46d3f22685684c30986f26bc98c2	0
zebra-chain/src/orchard/arbitrary.rs	5686	2eabb4db744661c2725262b2d6d3152b2f14b9b73f2e2e4438399389ce0341f4	0
zebra-chain/src/orchard/commitment.rs	9137	8179aae615fc05d61f0c62b42f5c399d695958bc3cd8058b25fda8b980cb7fe0	0
zebra-chain/src/orchard/keys.rs	7762	67d77ec4946b8439f69756fb2809de0394293957613a803f64b51c2342ff8897	0
zebra-chain/src/orchard/note/arbitrary.rs	857	6bee4793cf9dfbc26d2067f79b7e6e13342987045cd2bf77d968603e0d5fcd01	0
zebra-chain/src/orchard/note/ciphertexts.rs	4023	88a1b4e069c8f84ef33fff3d4dcd39fed39f86f81aa0ad305afeeeb330b56c18	0
zebra-chain/src/orchard/note/nullifiers.rs	1162	03df8a232a1b86e118e7556434d7488476f94c66ab15decf86852fe0ca05566b	0
zebra-chain/src/orchard/note.rs	3848	0e66b1742e47bcec0e13659be92ce4403c6ddf1372105ccbe9abc0699401f9d9	0
zebra-chain/src/orchard/shielded_data.rs	16997	c4873e6049ab863eccb08538f6762795b8e027bde737af269eacf37d2314baa7	0
zebra-chain/src/orchard/sinsemilla.rs	3726	cea1cfff25c7bb20799c38cf934576d5190228db0a840f21307be0f83ee6815f	0
zebra-chain/src/orchard/tree/legacy.rs	4600	258771be5efaae5745acc0db5e4930f67e38bd5896a00e522e5ab65487ef777d	0
zebra-chain/src/orchard/tree.rs	24324	5d583db4505ee612894a01cf01a97b61d321f09615a7a3f2ec56d42006e800ad	0
zebra-chain/src/orchard.rs	598	b7f0abb3cbf0981292a87835c0eea78ac2be639a5437024e5d07f03d7b26665f	0
zebra-chain/src/parallel/tree.rs	11332	5ba43594a39938e06775bdd78431040d3d81690988a2a06e14e26e667d58a810	0
zebra-chain/src/parallel.rs	50	ebbfb209c4f369a11112a321b5cfd9a6aef5bfcbb432b6d5e5eca7609427f16d	0
zebra-chain/src/parameters/arbitrary.rs	2132	f37328224e67d0dd960b2f65cfe02c38caa37ab4cbe10cc5c698fc04733d06e9	0

zebra-chain/src/parameters/checkpoint/constants.rs	1012	b2bb552020f2b36e6cad1f167866046275a47a37c430f4ebb745ad1981354ab1	0
zebra-chain/src/parameters/checkpoint/list.rs	9141	e46116db16c16dc26495836cc5193ffe4155bbe80693f8297c08321c47811046	0
zebra-chain/src/parameters/checkpoint.rs	132	d97706dd4dba16c13c9dc94b127bb0dcc2963d2f03181b195957878b1bc83147	0
zebra-chain/src/parameters/constants.rs	5172	cee7b0c58f32f01852f17aa8241a25c78bffd408844fd1c28e8d616af9f74634	0
zebra-chain/src/parameters/genesis.rs	330	ac7ed04e2281411a4871bdc40c64b66055a3e348f492d86adf26e5ece0774e5f	0
zebra-chain/src/parameters/network/error.rs	5052	e4a8d3b842f54e59a6cf51123f960570c8bbcdff30bdda97f73bb3819501ef50	0
zebra-chain/src/parameters/network/magic.rs	1485	4b7388c7671c53479fef36495c186a0c900a0a1cfe39196b29150ba7c351964f	0
zebra-chain/src/parameters/network/subsidy/check.rs	12041	e54ea88ec6c2eee7012f73b77deaa189de9998591fd17d9da88327e9e976d62d	0
zebra-chain/src/parameters/network/subsidy/constants/mainnet.rs	11820	f6869352ac82da7f44f09f9d266c5e7060de4e63b1f34dab4d5d7abd27d4461b	0
zebra-chain/src/parameters/network/subsidy/constants/testnet.rs	11177	30640db066a2fa7ac23fe8e20a072560306eb1f02a242887c00609a4a52772ea	0
zebra-chain/src/parameters/network/subsidy/constants.rs	2843	cbfca164c313ae2e4f1a53b39c804be08a1e915ad43015aecfc2aee60c8a49ad	0
zebra-chain/src/parameters/network/subsidy.rs	43747	6e17a1e16929b92798fd69f9118f10575a66714388588c6bf13d23ca8bf44e49	0
zebra-chain/src/parameters/network/testnet.rs	64246	3a53e4fe036c2dd982461c0450d1b5f3e6e8e8d87c96fbecb5a6894b0dc36979	0
zebra-chain/src/parameters/network.rs	17334	e4d766dec1b9cf7eeca712b82763d4e33d32b83b94793e16d7d2c07abd35a52c	0
zebra-chain/src/parameters/network_upgrade.rs	25183	87c4eaa567f370c8abcee6c3ff1ce8ec273aa28f423df0245006c3fdd9ea294e	0
zebra-chain/src/parameters/transaction.rs	627	f0b3473484b99c45fd2b4d9dc66d47df07d8924f28bed95d429bd9ea2f0de59c	0
zebra-chain/src/parameters.rs	958	23803edfe51d24ac0ee1986111a9fdd5bfcf01c602173984d35135fac0243970	0
zebra-chain/src/primitives/address.rs	6945	01f3e586d7c4474d6830b6e38c6062b46fd5beee98636a8c44cd328f459b368e	0
zebra-chain/src/primitives/byte_array.rs	591	773e9af39b36d8f5e14420a17d216aa8a233ca411644c030af9144c535887020	0
zebra-chain/src/primitives/proofs/bctv14.rs	1853	1c47358444f620a39dfb63b0f568e61da03c0332dac7c7d47a31b8af8522e028	0
zebra-chain/src/primitives/proofs/groth16.rs	2089	18533ba477cd2f05b647fa47bcd97973d77b6544263fa63d39783db0a6075d1e	0

zebra-chain/src/primitives/proofs/halo2.rs	1585	8d099e24b371dd6f71ccba106cdec0d114b3acf0d383573586d972d7fd77162e	0
zebra-chain/src/primitives/proofs.rs	773	06c9bbb9a03f9043302e7a3aee5bc52271bcb9be54cd5f31508a4f656a1b87f3	0
zebra-chain/src/primitives/zcash_history.rs	13671	5afc3d83e3ac22d89d32a8a36f69fca175ef7c515257b2c2300dfdbc32542c79	0
zebra-chain/src/primitives/zcash_note_encryption.rs	2715	7116a10cac1a7967d8b7370f700ae4f9e1b74044b663fe92644516c64ebe0543	0
zebra-chain/src/primitives/zcash_primitives.rs	20361	367fb2a554dc6fe631176e15a088f3f82fa78b27f184f2bf98fff79e62cf5741	0
zebra-chain/src/primitives.rs	604	ac08248210a1b0715d3addffbbceefb58af2364f902a00b1b24b4167c81c4340	0
zebra-chain/src/sapling/arbitrary.rs	4816	3e1e6cb51df6f1d628b475fc3eac6ec4addaafb4ae8c4c0c1b1e957ac8d35778	0
zebra-chain/src/sapling/commitment.rs	4339	d0317def005af6256855210d70c74125cf4ecc58bcf6ea44f9012c13c6c8fedc	0
zebra-chain/src/sapling/keys.rs	12808	03b4a8a0d9eaf0c324eea219e1ee0488a1d8359507289590f6bc7ab856e2ac32	0
zebra-chain/src/sapling/note/arbitrary.rs	857	6bee4793cf9dfbc26d2067f79b7e6e13342987045cd2bf77d968603e0d5fcd01	0
zebra-chain/src/sapling/note/ciphertexts.rs	3903	a2aba8d43f997995a7f43e017cbbe1063e71c21f943828e297d899c3b878f3d3	0
zebra-chain/src/sapling/note/nullifiers.rs	1646	cae53d6498a61d058c58a60ce7c3e7e51585556e31fba7b3304a6109dde41da9	0
zebra-chain/src/sapling/note.rs	1097	45b7021ec07f8a5f1563f84690a7ae7ec067f65a675ec02e794fc443d7cc5703	0
zebra-chain/src/sapling/output.rs	13221	ba0547370f0283c4c3112a76ad83d833b0f82a2120c470ad4bbe3f46495ecbf4	0
zebra-chain/src/sapling/shielded_data.rs	13922	504e5a9aaa55839aea6d2c6dde024a27932e999c794ff89e5f6e767923a183e5	0
zebra-chain/src/sapling/spend.rs	14770	957b2cb4a712a0e7c3e0931642b557afa0f7885e3e80ae2b8eb4a962cedd1931	0
zebra-chain/src/sapling/tree/legacy.rs	5391	8d13e720180681dd31955acec27207a13f668ca964ae81d150fe2b996a89e08a	0
zebra-chain/src/sapling/tree.rs	18270	d4c4b3593b1ffdfbf8cf97ee9d374970e5b8dc8242aa27d999409d7fa668ede9	0
zebra-chain/src/sapling.rs	1087	44ae19da15e35988abbfa94f93a2023f869188cff52a55f22523395f36d6b4d1	0
zebra-chain/src/serialization/arbitrary.rs	4595	4b041e61cfb03b854997a48f9095558b3fb24b94e628a4c3562646cdc0c2acc1	0
zebra-chain/src/serialization/compact_size.rs	12681	ee2d9fb79d02cd62c0f956442e18ee57201f239b6aa66f5193eb9857d9e33706	0
zebra-chain/src/serialization/constraint.rs	1828	7205b3b6d684ae47b18bd4ef31bdf02e98b94ddb4d49244c3770218443063f19	0
zebra-chain/src/serialization/date_time.rs	12483	d506fd40b83e2973c895ec36d27235c809bc6577813365434d23c4524267e8b2	0
zebra-chain/src/serialization/display_order.rs	1579	6f9eef415a1e46bf2a2847466e65feeb59d8e2d60efe8552577713fa03fda8aa	0

zebra-chain/src/serialization/error.rs	4875	ebd54eca9cad08f85f0f0c079f347d1981a7c6f3e9419d79c0ceb97ef66b1c57	0
zebra-chain/src/serialization/read_zcash.rs	1168	62670e1401bdca94f4d9f6cd2b6c3c85c7584dcbb3831379a7d33429e50baf40	0
zebra-chain/src/serialization/serde_helpers.rs	3152	8c99449a279a4ebfbbec19e29db59948d4b3a5372027b31fe0645771f2c2028e	0
zebra-chain/src/serialization/sha256d.rs	2063	de768bf81066e297f7f63f4cfa3d17c9d3af882a27867c5e8c4b9055ed99f830	0
zebra-chain/src/serialization/write_zcash.rs	662	4efa398b718d08fd2eaadfb532243c17c43fe488fbe0ce3d52049de3982dabb3	0
zebra-chain/src/serialization/zcash_deserialize.rs	10532	14b4c9824995eca9cefbc58cb3d44934957eff97673f930c1101eb27c60172fb	0
zebra-chain/src/serialization/zcash_serialize.rs	7108	55c9faa8ecd0891ca27af3d89f246d0d5fdd6988e7d9fcc56004909bac40eb97	0
zebra-chain/src/serialization.rs	1525	ebbbed84ff65f46a705f687d5760572444e76d9fea97c70e8815e00eda4ae41f	0
zebra-chain/src/shutdown.rs	928	01b43125129536b7d964368d5a40dae892bd9bb74b456eead8afc0540df70a10	0
zebra-chain/src/sprout/arbitrary.rs	1764	f60e27780ae48066f427d32c054ae0d14eeb341a101987799ee3a3a03cbfd0d0	0
zebra-chain/src/sprout/commitment.rs	1959	b5cd52f35dd286b4443d748f68d0a5ca6ba6a1f5e1f1f053fc0aac8d5b048319	0
zebra-chain/src/sprout/joinsplit.rs	12408	225314a0c4d61733ecda0df7371e4d1402c4d9dc5c3c5bcf61607d1551a50cc9	0
zebra-chain/src/sprout/keys.rs	920	ae215808e68ef32bde73763b47088ca8e1b152ef536a841649f6f4781283a341	0
zebra-chain/src/sprout/note/arbitrary.rs	451	69dee95b10e0716bb56fcd3bf6afcbd88c094b46eab666a3fc430941ba805675	0
zebra-chain/src/sprout/note/ciphertexts.rs	1871	78084e824e1e3a149c1ce3915b0643e05178e33981f3f17286b423dbdfa81ff3	0
zebra-chain/src/sprout/note/mac.rs	1505	a21e86a1105814f54ad84a92574b86698165c6dd0d1d7ed4cabedd2dafd0415f	0
zebra-chain/src/sprout/note/nullifiers.rs	1551	0d64e64a4822b3ddd1725a0c0b287964104e7d13e5023a048cfec614d04f97d7	0
zebra-chain/src/sprout/note.rs	1166	78660a4ac0863cda459e07c7c75b00fda43d622292382064a2cd4413e5495191	0
zebra-chain/src/sprout/tree/legacy.rs	4551	4032a5bc7c51e428314399c8334015176847871b722271887585703e48ada6b5	0
zebra-chain/src/sprout/tree.rs	13981	996c2fd30b1ca8082bf871119305068d14cdd04eec9c7d3df5894eb496ff94cc	0
zebra-chain/src/sprout.rs	364	6bbede02b43bbf7a935458e4b1703f4e74218e41f4384d287d7473dd59b3df6f	0
zebra-chain/src/subtree.rs	3886	4dbe0b16ea003164a1d1f48b218baf494117836b1fdc5eead9814363fac906a8	0
zebra-chain/src/transaction/arbitrary/shielded.rs	21374	da2dcb7e899c41a867fc382a4f96565ef72282086ca758066f3f51649aafb4c4	0
zebra-chain/src/transaction/arbitrary.rs	40697	140024574bacd582d1cb281a3671983040423e6557ba740f87359cdbf24fce1b	0
zebra-chain/src/transaction/auth_digest.rs	3496	1e6ce46a66da8d4d35e18b9c220d9aaef74a34dbbdeb994bf971d813a1a8718f	0

zebra-chain/src/transaction/compat.rs	14448	ddf86281e59c80e95b1ec3edd3af7ec65d8ff5cc5f7164c2febf5be8a10ae109	0
zebra-chain/src/transaction/hash.rs	9968	c25708575e6843b699facbe77784cd7b2421cfe7ebdfa0fe95a050b4068fec20	0
zebra-chain/src/transaction/joinsplit.rs	4891	155b6fd2368259208ed8d92f9e86cc6a206fd5deca6b827b6be3a73a4d3a9875	0
zebra-chain/src/transaction/lock_time.rs	5099	dae806f6fda11960bdd9057da307f62155f5385bdc134388b343187820745544	0
zebra-chain/src/transaction/memo.rs	4615	08a875626482ceaf141f1f0c5799822949153051826 9b57f9d02eebd858e72c5	0
zebra-chain/src/transaction/serialize.rs	32372	7fceb04ef65c4829fbcb3ce253280d34ea6fe64817e2e4a65f3b6e775ef0ab2c	0
zebra-chain/src/transaction/sighash.rs	5568	3db6896163dbdb4a0c88e788d3d9e2ec29374e0f8900378c55a251f2f97da193	0
zebra-chain/src/transaction/unmined/zip317.rs	10256	f9963948c37ad2a23e2febdfe0eac6d7d5f3b663d8dc412a9b6e841ba205f708	0
zebra-chain/src/transaction/unmined.rs	16261	f852ba4af9c812fa777c5a795caae250994946e7f09d69a221d0d0c98ecf9216	0
zebra-chain/src/transaction.rs	60832	8c24c5c78507be94b192690c46e4847da30f24b922013a9de872b6b1befdeb06	0
zebra-chain/src/transparent/address.rs	18822	7865566939e6b3222a12a534225bb2df5869aa4c6cbf5fdbf854208e004737c0	0
zebra-chain/src/transparent/arbitrary.rs	2638	9be4a12a5f30af236055a2ac52a1b95b844df6429e8a7685240dcade94c9ce6c	0
zebra-chain/src/transparent/keys.rs	846	88ba431df453bf5fe71627394e61f4246e4cc3f1e66944562a8df27009a506bc	0
zebra-chain/src/transparent/opcodes.rs	407	d45a159a18bb94c33394c54b2e2a4c3cec812702ebf6002e48cba16f13a12f32	0
zebra-chain/src/transparent/script.rs	3318	d85b97c2a244dc88a614e8170980be2cda7d6785a446b550b7642059daed6da9	0
zebra-chain/src/transparent/serialize.rs	11181	04a4b80a25ad61446c6a0bb94d843bca2d984dd099a4301b0551aa9ed660c0f9	0
zebra-chain/src/transparent/utxo.rs	8200	fb9a156731d924a0e41fb37c4903f50630eac0f304dfea9e4d00be4a3ba69714	0
zebra-chain/src/transparent.rs	14247	3db4148660472fd980c0650bb517fb0beebb8f2e49ccbc3b021a354243ac3c01	0
zebra-chain/src/value_balance/arbitrary.rs	1854	125e78fa73d41745a0824dedb3111bee24330ec3dda13c1a1d04a13712514583	0
zebra-chain/src/value_balance.rs	25713	f259fe4e9cb12149cd8b38cf900e8dd80e1c3de571f480912c1203eb3abb13d0	0
zebra-chain/src/work/arbitrary.rs	512	3b187c27d5a165cdca00f04c5484a15788c22a90863a75d0ec7ae3e98200bd17	0
zebra-chain/src/work/difficulty/arbitrary.rs	2535	6100459afb3105a3c4042ff14c7f365b5f8f1dd580ada704f640071d18636c39	0
zebra-chain/src/work/difficulty.rs	28204	82adf21c7a8d95d50f5cd1b73cc70b32e5b956f54f284a1bbdf6d31c1d1f4a59	0
zebra-chain/src/work/equihash.rs	11789	55f50d87fa325f94611b3b8b6f06e84f2c7ce475bc6334f86463353ff34639d5	0
zebra-chain/src/work/u256.rs	648	ac8d55d47e3e6a677bcccbc7144fa491fb45fb46d13882b344c043a9380013e0	0

zebra-chain/src/work.rs	168	09c782cfa1b6b6b997f5666cca8e54e2ef01d4ea01b2b8356f4aed1176b18ac5	0
zebra-consensus/Cargo.toml	3748	d723c9e8bc329f5929d138e6bbe87e4126a6784b50dd303ecf42ef1aceab855f	0
zebra-consensus/src/block/check.rs	11620	cecf89ce3c9b4b9498d79803b6bf63df85219c4eb6a0ffb45d60570cc5336d51	0
zebra-consensus/src/block/request.rs	990	8e608e56375e3dc238b9ad9e7a00c2bb57e636016011f22bfc96971f80ccf91f	0
zebra-consensus/src/block.rs	21360	44bb3b00a101ed03177c4add8a4db66bcf13e48170063d67acdebc730b9e9ab9	0
zebra-consensus/src/checkpoint/types.rs	4573	463874ec0d7e38cf6a7a14af6a96991b51ca2a8468c9cc7ee749956c4be98f8d	0
zebra-consensus/src/checkpoint.rs	57423	fc2586f730f2cead457ed0be072a98970e2d68e19a33aa94e858a85b6cf98d07	0
zebra-consensus/src/config.rs	2773	ba25240087fbdf66e96b92bd27134a005834d4e0b73e0a8ab601249dbc62fa30	0
zebra-consensus/src/error.rs	21408	b19e8f0bb9708a090e61f9083c509093913d25f96412ab2ac01d7a7dcd17c18a	0
zebra-consensus/src/lib.rs	2664	91568d6f45e413a5273a2e4a248f4e46daf5f08140de686522885bfb97532589	0
zebra-consensus/src/primitives/cache.rs	20322	8d6c0c47d7278672bea75ab3564276b174c8f2989253b54e8b083503d9a4b36c	0
zebra-consensus/src/primitives/ed25519.rs	8118	a3ce7906723003adde35d2ec0f1dd8bb797adf559aba6410d685e6ed7efb5aaa	0
zebra-consensus/src/primitives/groth16/params.rs	1151	f0746e2513991205ec706e041d43994219cc8f8765e19e9c1451533d9049d47d	0
zebra-consensus/src/primitives/groth16/vectors.rs	2746	67b0a451a9247c6e03fd812f010a1d71cb5588ffcb107c6fc50e4ee913e9986b	0
zebra-consensus/src/primitives/groth16.rs	6820	278bda853282f1e905f75dde83800f827cafc4677557f2ed3ca9933c0afc18f0	0
zebra-consensus/src/primitives/halo2.rs	28247	437f487a9a5240b1303448877fd2161aa6b3b0032782c3d4fab630c5b3b67c2c	0
zebra-consensus/src/primitives/redjubjub.rs	7934	c65f9bfa6d6de25aee537ada8fc08137b25b57661716cf077a02996d0cf9de02	0
zebra-consensus/src/primitives/redpallas.rs	8681	260cef69fe3728981e7108b9a3a6d6f16dde7512da8ec6fb2d9be86732131f92	0
zebra-consensus/src/primitives/sapling.rs	11298	486142019610ae247b0be4537f3b2eb73c388360c2fdaa70f700757b2966e5ef	0
zebra-consensus/src/primitives.rs	1498	a9659d944688bc47150e520ab223401083a78d247e9017f970b1bd70f4afb554	0
zebra-consensus/src/router/service_trait.rs	625	2bd2c1e22580d70ac0d70aaa7684bcf9167f1c67414ac440f906fc240bfb2c91	0
zebra-consensus/src/router.rs	18915	bc499c2357c9ad63bf32e8ccf0ec3df27d7e6a91a2c09225d06976f3e3e35bd0	0
zebra-consensus/src/script.rs	2220	e244b00fc1c084f5c48a8c07a56155d98805a843967571426b3596098cd204cc	0
zebra-consensus/src/transaction/check.rs	35390	14a3995600fe10344672ee7aba44fc87fc527cee758b60ff391235407b98154b	0
zebra-consensus/src/transaction.rs	63491	cb84a61366b275f76e144aff8fa2a374053741a567ee6dd5a070c75985424afe	0

zebra-network/Cargo.toml	3570	8a428d90481eb5a1eefae34b0eb0441d96613d94bdbaed7505692f66c5cf9995	0
zebra-network/src/address_book.rs	37540	0c4ca80d952f174b04374bb5201946474c7a6ae141ed83bec072ca0b334a7de8	0
zebra-network/src/address_book_peers/mock.rs	2599	d42e6f85abdcb2acfa65baa37a0f8586c141f7b30237328ac703220d228318d2	0
zebra-network/src/address_book_peers.rs	1184	55d724e4d4f6e15d35fa67fc88eb45b1dfe004c22b49eab50f415ae0f639c038	0
zebra-network/src/address_book_updater.rs	16551	e579c9b0cc6dcc572e4714eb6eb2737de730039d5acac4bc1c792a74cab90bc7	0
zebra-network/src/ban_list.rs	4247	d4530bac69640902bc74f366e4b1bab97bccc09156455f4acf0fa41819c1837a	0
zebra-network/src/config/cache_dir.rs	2334	2636a3b0ca50996bc77a6a90f49cdfa1e35b391692e8ed53cddf6aed4b01418b	0
zebra-network/src/config.rs	46088	915904806ecf15f43ba41498178ff3bd7d4632b58e226431ee7d076eb61d6435	0
zebra-network/src/connection_metrics.rs	16592	18ae40f2b8f01902efeba315fbc9417ec6a1fbc5fbe9083287719ec5e52daae0	0
zebra-network/src/constants.rs	26598	3d6366de5794f0ce33e9933830a27cf723fe4d5abee359cabd3b1c168f88f077	0
zebra-network/src/isolated/tor.rs	4908	a3ddad853b71e8a780edeb904ed9b1ebf5e7cb23e792ecc09c3c16d10aceb3d2	0
zebra-network/src/isolated.rs	5742	843cab4371ee5a146477a5dad755f1e0b11d0af73adcfabe77dfd3ddfa8f6c34	0
zebra-network/src/lib.rs	9059	881795f1aaa97d5130991d0833514f7e9043fc745103323236e3009d64afc3f8	0
zebra-network/src/meta_addr/arbitrary.rs	4067	916cc073daa359d4e1b11269721c34a4bc9af1c33bfbe60ff30a4693ebfa8a92	0
zebra-network/src/meta_addr/peer_addr.rs	2325	bf52563c0c38e7ca1065ce969abf41f728a5b55aa96858640eb14f2c48473a00	0
zebra-network/src/meta_addr.rs	55335	578598a8051f2fbb6db11b7cbdc1d0c986edc58f2fa7c921639cf59bf3037035	0
zebra-network/src/peer/client.rs	26617	71a7ac6fb96ada2c34af1e7a7189b08faf2d94a281c1ccbbed44c027caf84ae0	0
zebra-network/src/peer/connection/peer_tx.rs	2329	04ceafe3973143a92f43c2d21a8a52724e60fafdba69ed8cb2d599f14f6d186c	0
zebra-network/src/peer/connection.rs	86775	030a663cb31274a2d43e8c69bfafceb6c4786362e19ca54d001a7b0fab1b9b87	0
zebra-network/src/peer/connector.rs	3476	bdffba1f63a5acb063d212a8912f5b1275c8dcdd5f26f5574e5c0b3462f3a894	0
zebra-network/src/peer/error.rs	11294	562ad412dd784c815eb17d72033c53d6893d2ee8a090a89c6e7dc0243b26266e	0
zebra-network/src/peer/handshake.rs	62855	61148a6d2f06956aea0730b2f68a56d922a0047245a83c7ad91d5773790844d1	0
zebra-network/src/peer/load_tracked_client.rs	3309	096826fa1b70fd83ffcd0994226600146b9edf4a3e907f5f1b3dd7f2681b353c	0
zebra-network/src/peer/minimum_peer_version.rs	1766	e84e1ac97c27df881a0b985ca91dbb30219ae206ce3a81becd451a81b7bddb09	0
zebra-network/src/peer/priority.rs	5427	89c24936424cb7d706e0be4535aa34d86362a531a1131352ceba7928896b1da6	0
zebra-network/src/peer.rs	1132	c8fd05c3b1aa1ffd88706542e9f0e111689cf9c7eba7927bb3d77d7e5324e45c	0

zebra-network/src/peer_cache_updater.rs	3224	d0d73a836a2f16be7e5df862349638818659b4b185b48bc3e0add6e3894dcd2b	0
zebra-network/src/peer_set/candidate_set/crawl.rs	14148	95fb8fbdef0fe25499863e454b9931fad43181993e74b74ddb96e1f9f567bef7	0
zebra-network/src/peer_set/candidate_set/rate_limit.rs	10739	b145656b67564bbb1aabe8b169a5b480ae3b6df13e5d7bdcee5f895e687f7c12	0
zebra-network/src/peer_set/candidate_set.rs	12289	8e43158894c50e025b5083b4b6ef00765221ab4cc138dac27007b00fd8ced0a3	0
zebra-network/src/peer_set/initialize/recent_by_ip.rs	3501	d5e7fc5ec63e5fad4aa3f084e114b12cdef52a98cbb805d84a38188e4855b4c9	0
zebra-network/src/peer_set/initialize.rs	56355	4e5274859e28d08778a300b6b08ae221d80b27c84be6bf95d3038fc1d61f01db	0
zebra-network/src/peer_set/inventory_registry/update.rs	1121	807397cebe2dd765b201f2d3272668deecbed42b0b965754ba72421cd7c57e65	0
zebra-network/src/peer_set/inventory_registry.rs	20018	164fce04348567ebd30ba27a7e53bcfb452d223fee4d5a6f78febde9461e2238	0
zebra-network/src/peer_set/limit.rs	8287	ef54417e4d8f9337243c2aea6eb41104d4587dd27eecc8494729bebecea58b70	0
zebra-network/src/peer_set/service.rs	4173	ed613f79e668a1bd0b5f5e46976851f9607eea1152fc5b29de9ef040ef570e1e	0
zebra-network/src/peer_set/set.rs	99829	9cf13afb49bb795d633889f72eb6427e05b6515e6530287940312c1baef75542	0
zebra-network/src/peer_set/stall_tracker.rs	1770	e3abcc336324de2c141c0fb56b1729b45e859540cd6edf3110e682e33bcc7bf6	0
zebra-network/src/peer_set/unready_service.rs	3632	3bb430290a2716863150876a7382576cd75636fb91c8a19e45201414ef66dd78	0
zebra-network/src/peer_set.rs	7286	ddc54fc2f0eefd0f9ca1f91f43888458416948394b119a08d06c6fcd774cb721	0
zebra-network/src/policies.rs	2034	2b87c933ecc92f00c9d57c154c9a5d17c228872442ac42cb18bf697452cdebdb	0
zebra-network/src/protocol/external/addr/canonical.rs	3900	21785f2c73b6a501f763ae9f99a319413c3d8655a432751f7b449d9dbc011635	0
zebra-network/src/protocol/external/addr/in_version.rs	3495	d1df383a9adc5dc6df27743b96bf0b1a1c66a046e398c37e60a4c985f79881e3	0
zebra-network/src/protocol/external/addr/v1.rs	5351	5d67060fe9d72950c89cb022f9a11d6ef428e32632fcdfb81b607a3147f4e9de	0
zebra-network/src/protocol/external/addr/v2.rs	12580	4a805fcb71e65d24b4386026695302d16c5b36a5ff91ca948ca20330d0e09442	0
zebra-network/src/protocol/external/addr.rs	838	1503d33f9a825a63fb248425061e7c3ae3259f5d3a9aa7ae65867981e74e2bcd	0
zebra-network/src/protocol/external/arbitrary.rs	4086	51b20d867a54f435c7b31ff8b78b97a180faf7e770c5febb396920b338894fa9	0
zebra-network/src/protocol/external/codec.rs	33791	16af05828ad59fe29e7ab7171c2d2ffd0717b5820503560f363a0d4bd959e7aa	0

zebra-network/src/protocol/external/inv.rs	7873	dd872231f6f0483035bba0ff6375b2a1f22cbc30b879f6c8422a6c45770e7e6f	0
zebra-network/src/protocol/external/message.rs	20628	2088412c6f61a59f7363fdfb7f7437d3eb7d274fb6e18f1240210e61065c7f90	0
zebra-network/src/protocol/external/types.rs	8076	38a4036c037409b65e7771d6064f1a1ea0b277a43e3b9597e231ccac20130e32	0
zebra-network/src/protocol/external.rs	789	c1dac27487c69d78d79063fe410c1bfa0d2802c97d286dbff443c4a845d37a7b	0
zebra-network/src/protocol/internal/request.rs	15177	5c2c28d395fbef23b317fd656ac791afe17d2912beb1b67748fffaa11e29e18a	0
zebra-network/src/protocol/internal/response.rs	5960	51c8e9591fc5b8b8ce40df7081acb82a9a28de70935cfa13235825963b4da1b6	0
zebra-network/src/protocol/internal/response_status.rs	3073	b4998072c9eb8c7cf3870d4064e0f75e093f25bf3ad5971ac0aa81bf6d5f63e7	0
zebra-network/src/protocol/internal.rs	147	f518710a0001cc025b65ed95aaa4822ad1d25b9a741951d414b2d13bc8e69ab3	0
zebra-network/src/protocol/types.rs	196	44822b63b4b7eb4c917cad8a8b4fa284b223d17449d61225ea4a44f2d64fc970	0
zebra-network/src/protocol.rs	243	5a1d111f074080ea6f2aa7097fa5cae6e59a4d9328b54464cb10083d4fa5bf77	0
zebra-node-services/Cargo.toml	1631	b285520f28ee0ed043e1d4aec3f96b3ad8ab57565264b91955b6689fb5899be3	0
zebra-node-services/src/lib.rs	572	fb72ff78f17a7d9f42bebeb3b4f5d715e8d0a9235c8152bd4a42606e7edf0689	0
zebra-node-services/src/mempool/gossip.rs	1091	fdb0f24298fdc4c19c79532b10bb3b2a4ad8df360b6fda9dbabaccb5778ec49a	0
zebra-node-services/src/mempool/mempool_change.rs	3200	7f1472a544f286f120d5d7788d9cd8011d82bfcdbb338f220f6b609eab6ff27b	0
zebra-node-services/src/mempool/service_trait.rs	567	b246580f9f90cea8fc4ca0b4b51ff8d525eaf70546eb25c6ea66f74b5f2663c2	0
zebra-node-services/src/mempool/transaction_dependencies.rs	6431	b61684106f462af64fed36b17add84ad7896063e7a046ddddc089c2310732eed	0
zebra-node-services/src/mempool.rs	8437	e65205394d08dd6c7bd0a1800bb30583d67f919e85a17a8c8fafb119ac72baad	0
zebra-node-services/src/rpc_client.rs	10715	5e970858a4ffe20de69bd449e20f9549c657125cb06d51e3f3cc6a03f4bc1e25	0
zebra-node-services/src/service_traits.rs	681	88a4a73bdbec68de62d530dc8b618dc62167e2dd930b07f8c16b309f757ecad7	0
zebra-rpc/Cargo.toml	4503	d13812caab253b9984e1229dc347205ceb83c098462814bf229318e6e1eb96e7	0
zebra-rpc/build.rs	6074	56c959b4d2157692acdbabe696d8d40ac1fd71502338bf92c15b020d7d1baa1f	0
zebra-rpc/proto/__generated__/cash.z.wallet.sdk.rpc.rs	110589	de4c9bf91cbb4ebead4a0b88f09da398246b0c68910a5b905165d0a58da3c2b7	0
zebra-rpc/proto/__generated__/zebra.indexer.rpc.rs	29727	af2d02371d71162de13cb4754efddedecdffd5e3c17f7eaffe8450e5066ab094	0

zebra-rpc/proto/compact_formats.proto	4762	913dec2938b0301f41577ef5927734ba8af0e7ebefeb6588a2c62299875e0387	0
zebra-rpc/proto/indexer.proto	2914	b8395c6d0a676651da0434d1515733edf2840f0ae08809d353fccef6b84e80b5	0
zebra-rpc/proto/service.proto	10892	77454df4cfb67e1fdc66039a43ec0bf3d6f2fae7b6eb66bec11ac7d63bd22eac	0
zebra-rpc/src/client.rs	2403	39bc8c4103fe737b5a880beaa65dd3fd75b87f8c6c43877251e8b8b18f982d20	0
zebra-rpc/src/config/mining.rs	7832	9f79e938e006ca571e631d5f159f723b56c9c7720219fadc2f4578c77e7ec56c	0
zebra-rpc/src/config/rpc.rs	4866	0c127b3b0c691f1a621f667a85c09e427374810e2fbf665ace7352b38aceb68d	0
zebra-rpc/src/config.rs	62	e54857d5e854c77ca9b69c4060e44e7935577b6e0eb8c288f9c6ad536da0a89c	0
zebra-rpc/src/indexer/methods.rs	15589	6c74175740dfe0048f162f0f3e892ba0bcd383558f1336608569a25fce969759	0
zebra-rpc/src/indexer/server.rs	2878	0e377d5efcfa0c59c8e47b169d6059e4d7db444f8af44ff9b7f99e2a4ea186a5	0
zebra-rpc/src/indexer.rs	2630	af51c7ecae54ad330524ec76c8e54eecda6793ecf29c0f4bd77b5b7edefca9c2	0
zebra-rpc/src/lib.rs	610	8df5b30d41157599a400a55c2240da6aad08d8ad8003c5436ddb9f7757371fbd	0
zebra-rpc/src/lightwalletd/methods.rs	50411	9e4d55a8fa44115958fb83b131f1e43a1bbd38184034a665bef5564ff1f64cf2	0
zebra-rpc/src/lightwalletd/server.rs	3724	703c589ff746a83566b1e248f7d0469f1043d620f0dcdb5be5255c0767f8a02b	0
zebra-rpc/src/lightwalletd/tracing.rs	6674	06af1349ca6b94824d878f9b7704238bfbde0bc0718194341732ab99bcc7a777	0
zebra-rpc/src/lightwalletd.rs	6656	01c998ea7b9d20a437c7022a5747a35fe293dc713398ab3d69faf248b4654890	0
zebra-rpc/src/methods/hex_data.rs	352	36e347b9b0c10aa786084d02de1a09fe2cdb061537fcdc450ee34758b3f34197	0
zebra-rpc/src/methods/trees.rs	8112	dbdcaa31d92b06507adf36ea5bc6167595fe89b90d6aeba37ff3b95a6dd6b5af	0
zebra-rpc/src/methods/types/default_roots.rs	3958	f210e97f16e0164dfa9e7cb8caa7b65f042c4e0166d4bbfca4e00e00db8c8234	0
zebra-rpc/src/methods/types/get_block_template/constants.rs	2700	e3f546ffcfd51a36da2eca8cbeecbcb5e516e80d9a85f61df929333becf6a2a6	0
zebra-rpc/src/methods/types/get_block_template/parameters.rs	4231	7361c51ace781e9608f0365015247f7f43de237478c5e311ea9fe9a14366bf37	0
zebra-rpc/src/methods/types/get_block_template/precompute.rs	19702	a3cd0dcbdea5d841d70dc05446c9ec2383ca6caeaf6e39d3af92981f33705878	0
zebra-rpc/src/methods/types/get_block_template/proposal.rs	8495	b3a505b59cef0d085f9c4416a8bb97757d8f862a02acf91a7f8fa456bf89de15	0
zebra-rpc/src/methods/types/get_block_template/zip317.rs	19132	f0b2e3a856bb80f794d0c7010072787d2c559f7ab25c614ac0d1153376a9428d	0
zebra-rpc/src/methods/types/get_block_template.rs	40875	acff697fe4a82cfe2394eb3d0fb312e1e3f397d4de39c00ae7caff1c0a0c8b03	0
zebra-rpc/src/methods/types/get_blockchain_info.rs	5496	338236d6d26464953be896b91996b7f1e65f9550548dcb3fe70d6a699ab0186b	0
zebra-rpc/src/methods/types/get_mempool_info.rs	697	9a5cc253df43124520b84e3529fd16d746a495d989de73da43708ad34adc90b1	0

zebra-rpc/src/methods/types/get_mining_info.rs	1683	6d0823dc4a0cd83d46f64f250b9d8efaeb79d338c5c436956b387ad02c47050c	0
zebra-rpc/src/methods/types/get_raw_mempool.rs	4388	b469e45bfca190619736350163a6f22e08deeb6e138a922dc0b344eb0cba57ac	0
zebra-rpc/src/methods/types/get_standard_fee.rs	1824	88a128d2601d5853d836ff262758910122389c909333313b1d25f89ee6404a35	0
zebra-rpc/src/methods/types/long_poll.rs	12013	046be6efcc09b8e24c7535d74df829a2f4666d01b64820e14bf74d1e3d6ce157	0
zebra-rpc/src/methods/types/network_info.rs	2021	6cf18d1a8024adf4a7f65da84e5f365930ec9440ca6b67f55b861c221e9acbad	0
zebra-rpc/src/methods/types/peer_info.rs	3275	134bc60a847f74cdf0c2078f614425b89095ce8119f6216d4288bf42bb6c21a4	0
zebra-rpc/src/methods/types/submit_block.rs	4029	1ff39142dbba6b562aca9f91cc2a9ce3b1bb53436d6ba6f39ca07fba02fa6378	0
zebra-rpc/src/methods/types/subsidy.rs	3730	4c046d7c4fae5ed819db02f95fa3c8c5d454575d904e8635eac580ad130a4550	0
zebra-rpc/src/methods/types/transaction.rs	56642	b0d6338831d0dbd929a08f441b6d95fc6d24a1b9fa11e97f2e9f67a4985c0d8f	0
zebra-rpc/src/methods/types/unified_address.rs	945	0fd42df2db48e76c2dbdec97984d38ef7ba9027f237e26dfcb163a63ae20ad6a	0
zebra-rpc/src/methods/types/validate_address.rs	2578	129d0c7437c80e83d1cc00d0102e272ccc05348e29de8e9ad7ca17150f162c6b	0
zebra-rpc/src/methods/types/z_validate_address.rs	3786	a96423c7e994bf0f257225f66d401493936a3a856fb818edbd76b1b1e4055e42	0
zebra-rpc/src/methods/types/zec.rs	6706	7b7fbce7b2ed5da18be4f49fbfed4de08726ec0e57e22f3b0d4c88439e23aa0b	0
zebra-rpc/src/methods/types.rs	426	68140a90fe4cb0a1c07ccf200297790e3e4ad9970f9f7aad5a0d0b1a7f18b159	0
zebra-rpc/src/methods.rs	211166	eadbe226e9950590359847c7b640b8f75e84c80d097887f6fc805a1bee82559e	0
zebra-rpc/src/queue.rs	11436	3cb47ea763a62686c5d3521a38ea5cd17caaf3c73b6f6097f7544040d3fec4d7	0
zebra-rpc/src/server/cookie.rs	2403	27bbc9950d8bcf4205bb70855b524a6c50094c7701ce6b537b8bd95e9d42ccbf	0
zebra-rpc/src/server/error.rs	4427	8f0442b00c49678db3eed2d373b9c62b3e0fbf0f71f3b42e65f78ce418d95f21	0
zebra-rpc/src/server/http_request_compatibility.rs	13127	322d240680f9a084e834501e814e5b7494f929108dc58e38b60160e41980248c	0
zebra-rpc/src/server/rpc_call_compatibility.rs	6616	3821156f7d0b7a65297d9d82ed99c2e30802563175cba5dd01ef085cbc38932e	0
zebra-rpc/src/server/rpc_metrics.rs	4027	2a2ced4880a96e25f122595c3b73a1e78ea294c3555621ba7bfb963fb8e19c83	0
zebra-rpc/src/server/rpc_tracing.rs	3660	5a12bafac5019423ff20854407cb9755b325c271ecdedfc1a4b81394663ca3d9	0
zebra-rpc/src/server.rs	8462	d091d3e58859f87b225ec8fd7602ac7329018de95b6a1cebdccda672b3714efa	0
zebra-rpc/src/sync.rs	29922	8cd437859f39b271643cd7a0230ec79dbd78c28ec90ec8115f709ca7b1e059d6	0
zebra-script/Cargo.toml	1451	a7877eb8f872c67f2ce79f5ae936baf243a6a8992c0a5f97f9461b9f749b0cef	0

zebra-script/src/lib.rs	19139	8e3e24c4c68f464e628e25f722f68a6f5d7a8c37f4ea4562c8e060178e63fd97	0
zebra-state/Cargo.toml	3444	5b64a2d0c0fff829a720054b2d28caa2d7576da3c3909fee5d581c22ea754102	0
zebra-state/src/arbitrary.rs	3743	4385f83332b6ff6fc77ea3598b60b5544ab058c58c47117faccf67c2620c9628	0
zebra-state/src/config.rs	22511	e9b28f9ef9e0add64b337f1a2c2fb2450de73c2d5e2828e377ec7e9d00ad393b	0
zebra-state/src/constants.rs	6446	a355d3f1b07f3a82ca673d80deab163b22c479b45558f5ea8a584abc5aeb0f5c	0
zebra-state/src/error.rs	29378	15c39917660edc96a60a84cc33ef7d66a4b3708b7d4a6d06984c9be098aa184b	0
zebra-state/src/lib.rs	3454	c685d45f261106f61aed627db3f9f8d4d447731d3a58981fd3d88280cef6826c	0
zebra-state/src/request.rs	73952	819b801382a2fe1dfa3a1e1dd240be9832662779c96cc4f338935b3239443a0e	0
zebra-state/src/response.rs	27434	a18cc3f68271c6c1e959df2f9791f0d59f3ad32c35099066495420dd3577fa95	0
zebra-state/src/service/arbitrary.rs	9145	40d69854b7c961e248194fac9726bdea0716b7204bc421cd07341e943e1416ba	0
zebra-state/src/service/block_iter.rs	6661	a04b688c87babc2b397b51088d5c5a73c28aeab90c3b13d094f69a90666a1f0c	0
zebra-state/src/service/chain_tip.rs	26686	19318cb5a1de6743264aaea1267e4042f45d0e8112b923f4cf4fecbda3e066d3	0
zebra-state/src/service/check/anchors.rs	19417	6e4699c1c092aadfa6f75a67b5543792051eae0f45f3e4e7b5e99efbe6e6f72b	0
zebra-state/src/service/check/difficulty.rs	16820	7e21f95a20446bd5c719a0f5febceda95a41523589e0831222ec1d02a318b9ba	0
zebra-state/src/service/check/nullifier.rs	9082	749be77f30b15da3e68f4471d8994056e4518c0adacac9009f76adccc94a77a7	0
zebra-state/src/service/check/utxo.rs	11844	e165110675ff77ec5e4144527c63b564e43de69f568349bdf50e0dd2206306f5	0
zebra-state/src/service/check.rs	18459	5e7db52f0d578345d34f1ef1f1a8b65ca3d86ac4cb1dfee24848b173a8e3abdc	0
zebra-state/src/service/finalized_state/arbitrary.rs	1984	dac47f5015238073f5e19e2823150c552d8a173fb1c38482c165a10c54e7ca3d	0
zebra-state/src/service/finalized_state/column_family.rs	10664	16ba1885be079d656ae288444356b899a1f5c8272627ea0270e3e340ce8d80dd	0
zebra-state/src/service/finalized_state/disk_db.rs	67360	2e8195b747134db362606f9a0ca1431e997c81bbbef53c815564f2dcff081785	0
zebra-state/src/service/finalized_state/disk_format/block.rs	11557	c915f7db3d7e447fb8c682ce9460d88a5cf9a3c013ab1dbd0bf612352ff1222e	0
zebra-state/src/service/finalized_state/disk_format/chain.rs	8404	bc3a868af5b0c7c567f92f8fd57b08e7bec176b3a682ba36935f67c22627a8fd	0
zebra-state/src/service/finalized_state/disk_format/shielded.rs	6227	5be1b8b9f01b8f92544d699501f36c38837f5e4d768aaed03c44d85723765e73	0
zebra-state/src/service/finalized_state/disk_format/transparent.rs	32712	8f49aa674fae2967ed6eb741a6fd465f1194975dee63d59c43195d3910759b4f	0
zebra-state/src/service/finalized_state/disk_format/upgrade/add_ironwood_tree.rs	4058	129f367d5e30958aadf3b2a181b71b9aa903d4ecf87289b7246310728b0fbffa	0

zebra-state/src/service/finalized_state/disk_format/upgrade/add_subtrees.rs	36415	633e1a6cc2e31b2b6cd84fd0511006dd475787511816548e1353511f2cdfc8ab	0
zebra-state/src/service/finalized_state/disk_format/upgrade/block_info_and_address_received.rs	12654	a8c927868d47fc4403f52f06703190b14556736a0bf6623917843a40714d5238	0
zebra-state/src/service/finalized_state/disk_format/upgrade/cache_genesis_roots.rs	6316	358c1b99b73cf022bb032bfead1da247bf3e05a0f8be178b0d7213b6ff989dcf	0
zebra-state/src/service/finalized_state/disk_format/upgrade/drop_tx_locs_by_spends.rs	2399	f0ee95b702f14faf8c15d7bcd63e96169c40a1c5493e118fb4e2c6d6777f8b0c	0
zebra-state/src/service/finalized_state/disk_format/upgrade/fix_tree_key_type.rs	5316	2fe6d0b5b4ff4cd8cea2ffc4d2568e902fbac61a544f79c15799ba7816baf1c4	0
zebra-state/src/service/finalized_state/disk_format/upgrade/no_migration.rs	1243	631b361bf44bf5a8273943821d816f7cb94bc9812c5e6b60d5c59e8cbec16e1a	0
zebra-state/src/service/finalized_state/disk_format/upgrade/prune_trees.rs	5330	3386c552662ef780579749f012a6095eebfef1166ff3b3d7115fba457d6c0c92	0
zebra-state/src/service/finalized_state/disk_format/upgrade/track_tx_locs_by_spends.rs	3959	32288068ac736fb662d4bb1efe47efe197c6ac372ec511b1ffaef6aa6e8d559b	0
zebra-state/src/service/finalized_state/disk_format/upgrade/tree_keys_and_caches_upgrade.rs	1795	4c77ec574152a4911e501c644adbbf6655cfc98b3f85cc52fbf5a7214fa243f3	0
zebra-state/src/service/finalized_state/disk_format/upgrade.rs	34142	bfdca0019cf7fc9f849469f4d571d7d7b2189d66f17a5e81d441dc390d87f162	0
zebra-state/src/service/finalized_state/disk_format.rs	5344	d60a25b81ddddcd06478afeef7d7f80f53e33d8424ee5519b6869017e724965c	0
zebra-state/src/service/finalized_state/zebra_db/arbitrary.rs	2745	9a73a46fc0b4001f9da6525b5ffedd0dae4d795a733ec4cb3fadafc43af6f89c	0
zebra-state/src/service/finalized_state/zebra_db/block.rs	29154	0acff88f1145be9bbd7310f1af9ba079a51e9cb714f22e98f0dabe2076481af0	0
zebra-state/src/service/finalized_state/zebra_db/chain.rs	12542	fe325e785bdbbd0604cb15ba3387987688122c973ab2b1dd8ca80ce8ad9676ec	0
zebra-state/src/service/finalized_state/zebra_db/metrics.rs	4165	aeb82941b63d769625c5538ad77cb5164b6ecec35efd9c5f16ed3ac673f08b5a	0
zebra-state/src/service/finalized_state/zebra_db/shielded.rs	39719	70753fe5df387b775d4a67cf3ef37844b2ec6a854537f04f5e58c285643adf49	0

zebra-state/src/service/finalized_state/zebra_db/transparent.rs	37562	ef3e4244939bf6bc6b2d926473ea1aa15d7e971982790f8ebd25c35ae35f225a	0
zebra-state/src/service/finalized_state/zebra_db.rs	17407	2d4b60deba1582b2db7cdecb63ce7ad2ca6b57fab08b7d36df8457cfb160b271	0
zebra-state/src/service/finalized_state.rs	26964	e2058ef9a866ba28f1088d126cb8e08baeeb2747197650b0e2c4135c41b6e83a	0
zebra-state/src/service/non_finalized_state/backup.rs	12980	dfb6b760333bf2a7d0cf9335e99e9237b4e97e2a945ab51d3daccc032f546a77	0
zebra-state/src/service/non_finalized_state/chain/index.rs	12771	875806ff47630be0417e01afd0cb55f5703efb1316c638f30aed51a205d53cf0	0
zebra-state/src/service/non_finalized_state/chain.rs	99848	45a866fc6aa9a25b33b3f3d0fe9c9e7b6b342321c053fb98cc627510c2e652d9	0
zebra-state/src/service/non_finalized_state.rs	44387	ad078fbb3820508e7d3bfb73808079410eefc850defae6953c9735fb07034bb9	0
zebra-state/src/service/pending_utxos.rs	2447	e4e6bcb31580ecb4149bd73f6e29b4c25c26e851e505eed11903ad854a864c5c	0
zebra-state/src/service/queued_blocks.rs	18212	a72e3b8224116ade6e374046d8b1e9cb02e81604a770775faf5685e4e6e923cd	0
zebra-state/src/service/read/address/balance.rs	5446	d508b7f7cd3d6fb7072c285a24b63cd0a5cb686b45e69378f56a186e38a05bec	0
zebra-state/src/service/read/address/tx_id.rs	11271	3a64ace37573d494cf7ba05cb18bce81b9792992abc1c8b8a56d986455346d83	0
zebra-state/src/service/read/address/utxo.rs	22150	f5da3cfa0f19649587aec3894bb935837bbfe78feea1d922b6af6c07f06f2d48	0
zebra-state/src/service/read/address.rs	76	7322803accbf92cdda3465716747331dfe4ff312d5e114dec63a0a4b1682a07f	0
zebra-state/src/service/read/block.rs	16716	91a1a61d0efb0303ce9ee330e03450caf02f660fb249157290bc5f959d135efd	0
zebra-state/src/service/read/difficulty.rs	23109	d3cb7d54208bb38859f4d90d75a865e7f112097c953a8fc71c7ffd92e6193306	0
zebra-state/src/service/read/find.rs	24184	c71cac21cf69336d842f9134fc29702e576698aadbbbd4b80ffae8a7f5e7bf88	0
zebra-state/src/service/read/tree.rs	13003	2acc5833a4623a19535b72674ebf2d16b812213662d04e10b9b78100e6fd3e7e	0
zebra-state/src/service/read.rs	1967	c97d9377ed551ec9a1c5840269283ab660bd9657354770f86be0ac411839859b	0
zebra-state/src/service/traits.rs	844	45b3d542f053bd8051ba7a0a09c4bbcf1c52e085db3ccdd207d548dffe1ee20e	0
zebra-state/src/service/watch_receiver.rs	5221	d7b19a5e719c87955d199c47342f841b5debb9d1f12a5645fe48e5a59017f1a0	0
zebra-state/src/service/write.rs	21424	32d969a4deceeecf356111601ea53c6a769969b8fd501a94f4d799e78325e958	0
zebra-state/src/service.rs	89598	e55de008aabc58d32acd2aa5db926a4b8a1abad25785e635354b72410d3b1046	0
zebra-utils/Cargo.toml	2479	4cd13d8311ff49ef02b5bbe0a4bf173206a877620d18268ac1cc74fed3e1d901	0

zebra-utils/src/bin/block-template-to-proposal/args.rs	1124	292747763d9a0cac875df0db80332121dca5196023ec15412052b559d6ea307a	0
zebra-utils/src/bin/block-template-to-proposal/main.rs	3341	acd70253ff5be567fb201bd1dcf9c22ea67b6356850e3c57dbf1300e7881eb37	0
zebra-utils/src/bin/search-issue-refs/main.rs	11383	4bf797fa77f1aa361f1b1c0961c75fb430323a724391722226a2c2c3b4a221c1	0
zebra-utils/src/bin/zebra-checkpoints/args.rs	4235	82749ca2aff21f997852631d2f52eacc50f45b363703d893e181f757929031dc	0
zebra-utils/src/bin/zebra-checkpoints/main.rs	10779	e12afab038dd4d963a578c7f13363e41474b8c7b35cd854b37ff9db342a42f5d	0
zebra-utils/src/lib.rs	569	57751376f2086415d8daf5d91199b1b6acea7db7aa965a7a656a3958dd0ee319	0
zebrad/Cargo.toml	11898	5009e65e19acd07abc76242e6985e58af12392f4212cd8a0fd2a43949321c673	0
zebrad/build.rs	2801	bf2e6e5ed85c550e066f76d09358ba6556ee9888b325eaa0f795966900e70093	0
zebrad/src/application.rs	24460	e8c626c846beb2a7d0ddc7fe4c62e536ce4221a18535f1f809aeee2abf837f0b	0
zebrad/src/bin/zebrad/main.rs	615	b4a238827f3c997a0480f55ca11091cf38bd96c10a15230fac179b7f1da5497a	0
zebrad/src/commands/copy_state.rs	16366	a0f607a11e34724f69b00cd48ffb0128d5ccd9d249c463762d774143ca8640b1	0
zebrad/src/commands/entry_point.rs	3919	ffcec92681b4fc230be1f57caa5e9a5f2da1ff850f5a305220fbf6d63345bf4f	0
zebrad/src/commands/generate.rs	3607	4fe97463036fb18c5c1eacb957d94a7a216517bf61e6a28f53c8abe41c22257f	0
zebrad/src/commands/start.rs	50122	09f04051904bd18e5b9d48fbd3b8a10eb1368b09d902593fa1fdf719a95b43ed	0
zebrad/src/commands/tip_height.rs	2075	2b081aac7ecb4216fa309092e165920e2418425048f13f1b276d66c85eb4a647	0
zebrad/src/commands.rs	4461	a1226fcce74583a5b315f6a17eb15397a96cd490a9cac5864e7f4b69e46a0e39	0
zebrad/src/components/health/config.rs	1414	f6167e6458048f08b2ad95d0b49126d5475d8a8765a3342c3f1062acb3f8345a	0
zebrad/src/components/health.rs	12857	479a0f58078467c0e7915023852303922d844dc72a5928aa78fa60bb5493bbd6	0
zebrad/src/components/inbound/cached_peer_addr_response.rs	3718	5a64415750b63e0e31ab7d95ed5707557c3d4edd666fa16c60d92792f57384d3	0
zebrad/src/components/inbound/downloads.rs	24041	e98333a03cad20d8097dc59fde005e296877dfc55ff361976121c895049e7224	0
zebrad/src/components/inbound.rs	30028	e298cc9545b1c65950d500e58bc7cfbbbb9461c86555c8129686e10bf4630fbc	0
zebrad/src/components/mempool/config.rs	3115	cf47f56cd0540fd0aae98d5a6b9049903a7770b428fc14e9067143e53cdbbac4	0
zebrad/src/components/mempool/crawler.rs	10379	6738235e04f548f3e1dde890ed42a958fdadac681c420c208d8c56dba62a0dc1	0
zebrad/src/components/mempool/downloads.rs	27739	725e13e314d39f9671be2d7d361bb23d5c7d84573fc3edfe46b236ba03ba03f4	0
zebrad/src/components/mempool/error.rs	3362	fc2a8f1602a7ad6b8f39eac699938685deda9dcaf5e722473e81e39a0d4b7d46	0

zebrad/src/components/mempool/gossip.rs	4912	c2918e119119042163ac700837d0fd015dd55ec7aa6c08ba1bb96265afb30efe	0
zebrad/src/components/mempool/pending_outputs.rs	2309	0f35179be39c78ff5cfbed9807c4a65bb28f8513fa53439c0b5268d02fb7e51a	0
zebrad/src/components/mempool/queue_checker.rs	2896	9dea92e21ed3ed08842d055725b4ad44ca8ca90059ce69bcf725734e0126a6e6	0
zebrad/src/components/mempool/storage/eviction_list.rs	5492	cd2e406b864977498110f1d9a78b5ea089cf59d6e963873fa366832f22660fb6	0
zebrad/src/components/mempool/storage/policy.rs	14196	89fdd3a1723f56d15848ea5b4836cc0153d69d33a54821fae565496e4721f45e	0
zebrad/src/components/mempool/storage/verified_set.rs	19581	1d2355c61bc7cd504bcae3704ea218a1f7f5828dc9ae544c40ec224063a9b7d7	0
zebrad/src/components/mempool/storage.rs	39586	608fe5e3849a0cbc8d214661d147c92e0c67038206c69ab5596bf5a4b17c227c	0
zebrad/src/components/mempool.rs	48032	f4d5a3c3e984201e3f89788de2e922a5d93e42ea6f61dea13e434d64de39467d	0
zebrad/src/components/metrics.rs	2775	e5f414f200ff61ac6b7e43ffd798848c830c8fdb2f5d57d5d274d27a4e805d64	0
zebrad/src/components/miner.rs	22597	8929f63db7eb43e30d416c52b61b34f9749ba4bc611fdc6d56a8675187fc75ed	0
zebrad/src/components/notify.rs	7218	37581938e8cd8cd279c799278ad711266cf4566cde1417bbc524593e41fcf9c9	0
zebrad/src/components/sync/downloads.rs	30750	876a9233e37317b7a07fc29067763766c1f1659c9d2ad9511e0151665bdd671c	0
zebrad/src/components/sync/end_of_support.rs	4448	51d368743189e923cfbb506e04b5f84b1b5aa04ce8175606a985ed8b9a3a9875	0
zebrad/src/components/sync/gossip.rs	9570	53d2494a541b11685b8f8359fb29e71af484aaa6d3d2dee33e0c08040a2774f0	0
zebrad/src/components/sync/hedge.rs	7930	20bb9eb249654b096888376c28b97c886e15cfe7d8d39d70d203254fee4daf8c	0
zebrad/src/components/sync/progress.rs	17110	413575e4caa05f3319ac7d866bdd22c18370021cc0837f7be8af4af4202d6ca1	0
zebrad/src/components/sync/recent_sync_lengths.rs	3461	c914e79f9004b0c00ab7b55098845e60de8eb2b50cd175b441e6823c2cdecb57	0
zebrad/src/components/sync/status/mock.rs	985	3d00fc970c96832beb0bc9269494692980639e55ef44d6f4cada5f9fe624c440	0
zebrad/src/components/sync/status.rs	3302	dea341fd74e3702ff529e96e46322eae4cfb46fe0dde9440780b3bcb3d010e89	0
zebrad/src/components/sync.rs	69320	cd8dbc4c6a466d4905b8b56b44676efbbf670d72133c7b38a97baeddbd6fa260	0
zebrad/src/components/tokio.rs	5744	664a837a178bd3d8f9800aaafb55eb7584f3e86530de46f3998e1a4bba0630cf	0
zebrad/src/components/tracing/component.rs	20990	606674e14b908fee082c838e4fc2c0db081fa23c393d649879f68aeb8ee62d11	0
zebrad/src/components/tracing/endpoint.rs	5790	041160c0d76ca1ac93924d214b2ead8ebe021d85af2d81900b96102c7e46bbde	0

zebrad/src/components/tracing/flame.rs	1284	0f405afc0d1d047e2673f5f7461ebcef22f72100209e8079a7cac768ea5cef70	0
zebrad/src/components/tracing/otel.rs	3877	c701aa2660b3036ad8644d9cfc5617a7ce89548206bb5ef98f6dfa7a7eb69aac	0
zebrad/src/components/tracing.rs	11997	58a5e65a0625cfc4ea48d4e809005bea39bb0a915ea8668f581a925f179a0c29	0
zebrad/src/components/zcashd_compat/config.rs	11346	25491271b3035609008bbc38a4232d61795e86bab316066af0cedf904b72d6bd	0
zebrad/src/components/zcashd_compat/datadir.rs	21111	4d5e76b865d078fe3380b71ecc841db52e7e6b9b07e632dfd6321b4f8579627a	0
zebrad/src/components/zcashd_compat/managed.rs	31634	49205c41376f20657cc3a22a33c7f0fc5bf32fbe6ce14c66e96bc42d191ef948	0
zebrad/src/components/zcashd_compat/manifest.rs	3720	0f65a7929bff4a42f02e79562389524137ee0ffd4c7e673e7def3b90ae1c188e	0
zebrad/src/components/zcashd_compat/preflight.rs	49050	367a84fcd608aa60318b05e32fb15f0bae97fe61a469fa14e364335f8203fd2a	0
zebrad/src/components/zcashd_compat/supervisor.rs	39059	45af07d79069d559c694f628a70a12678542da82bc0d2fa69d864ccd08d39311	0
zebrad/src/components/zcashd_compat.rs	919	9915d1f955a02c290e4bacfbc1159cecb2fbf07f2969bfbb6c02708537da8317	0
zebrad/src/components.rs	838	c17d1c9751ee8e18b042702b60af0a494630ffa291728d9e78f1dffca7328043	0
zebrad/src/config.rs	8478	f855468bf1695ae8f653573bafb9da3d97a4daaa75c9678be5c37915526542b3	0
zebrad/src/lib.rs	7582	1dc46a83cc05319f39a9db483964f3193f877b356caf6654da36bd8f0e57646f	0
zebrad/src/prelude.rs	307	02129974f3ab542a5807384e0ac63a5c00681fbfec9fe3349028c78e7b9b6886	0
zebrad/src/sentry.rs	34587	d08e1a7cd0dbd01c61934c391fc0cb85aa9038a0e2316b2c0bc06872ea3f2ca8	0